Geopolitical Shock Transmission in Emerging Bond Markets: Evidence from the 2022 Russia-Ukraine War

by

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ABSTRACT

Geopolitical Shock Transmission in Emerging Bond Markets: Evidence from the 2022 Russia-Ukraine War

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This paper exploits Russia's invasion of Ukraine on February 24, 2022 as a natural experiment to examine whether geopolitical shocks alter the structure of credit risk transmission in emerging bond markets. Using monthly data from 2010 to 2024, we study corporate-government bond yield spreads across three maturities in Kazakhstan and Russia two regionally linked, commodity-dependent economies that differ in their direct exposure to the shock, providing a natural comparison design. Our empirical strategy combines three complementary methods.

First, an Autoregressive Distributed Lag (ARDL) regression identifies contemporaneous spread determinants, finding that domestic liquidity conditions dominate short-term maturities while global factors U.S. Treasury yields and oil prices shape longer-term spreads.

Second, a Markov regime switching model, estimated without imposing any a priori breakpoint, endogenously identifies a structural regime shift in [MONTH] 2022, validating the invasion as a genuine break in bond market dynamics rather than a continuation of pre-existing trends.

Third, Vector Autoregression impulse response functions, estimated separately on pre- and post-war subsamples, reveal that a one standard deviation oil price shock [NARROWED/WIDENED] Kazakhstan's 1-year spread by [X] basis points within two months pre-war, while this effect [STRENGTHENED/WEAKENED/REVERSED] post-February 2022, indicating a structural decoupling of

commodity fundamentals from credit risk pricing. Across both countries, the post-war period is characterized by [DESCRIBE KEY REGIME CHANGE].

These findings contribute to the nascent literature on geopolitical risk transmission in emerging fixed-income markets, with direct implications for investors who historically used commodity price movements as leading indicators for Central Asian bond markets.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Dr. Denis Pascal de Crombrugghe for his guidance, rigorous feedback, and intellectual support throughout the development of this research. His expertise in econometrics and dedication to scholarly precision significantly shaped the analytical framework and methodological choices presented in this paper.

I am also grateful to the institutions that made this research possible through the provision of data. Corporate and government bond yield data were sourced from the Kazakhstan Stock Exchange (KASE) and the Moscow Exchange (MOEX). Macroeconomic and reserve data were obtained from the National Bank of Kazakhstan and the Central Bank of Russia. Additional financial and commodity data were accessed via Bloomberg, Refinitiv, and the Federal Reserve Bank of St. Louis Economic Data platform (FRED). The quality and accessibility of these data sources were essential to the empirical analysis conducted in this study.

Contents

LIST OF FIGURES

LIST OF TABLES

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Chapter 1

Data Sources and Pre-Processing

1.1 Data Sources

All raw series were obtained directly from primary sources: national central banks, stock exchanges, national statistical agencies, and FRED. Table 1.2 lists all 35 base variables with their country, source, and monthly aggregation rule; full source URLs are retained in the accompanying data dictionary file for auditability but are omitted from the table below for readability.

Initial date-format standardization and daily-to-monthly aggregation were performed prior to import into the analysis pipeline, rather than programmatically within it. This was necessary because several source files used non-standard and provider-specific date formats (day-month-year, month-day-year, or year-day-month, depending on the exporting agency) that required manual inspection before they could be parsed safely; performing this step once, upstream, keeps the analysis code focused on estimation rather than repeated ad-hoc date-parsing logic.

1.2 Aggregation and Temporal Disaggregation

Table 1.1: Monthly aggregation rule by variable type

Variable type	Aggregation rule
Yields, interbank rates	Monthly average of daily values
FX rates, stock/index levels	Monthly average of daily values
Sovereign curve parameters (NSS, ZCYC)	As published, or monthly average where sourced at daily frequency
Quarterly stock/flow variables	Disaggregated to monthly (below)
Already-monthly series	Used as published

Two variables required disaggregation from quarterly to monthly frequency. **GDP** (Kazakhstan and Russia) was disaggregated using a related-indicator (pro-rata) method: each month's share of the corresponding quarter's industrial production index was used to distribute the quarterly GDP total across its three constituent months, following the standard related-indicator approach to temporal disaggregation (cf. Denton/Chow-Lin). No boundary-smoothing adjustment was applied across quarter breaks. **External debt** (Kazakhstan) was disaggregated via linear interpolation between quarterly endpoints, since no suitable higher-frequency related indicator was available and the series trends monotonically over the sample; Russia's external debt series was already available at monthly frequency from its source and required no disaggregation.

1.3 Data Dictionary

1.4 Series Import and Validation

Each series was distributed across source-specific spreadsheet files, one per variable, each already brought to monthly frequency at the Excel stage (Section 1.2). A format-detecting reader was used at import to accommodate three distinct file types encountered across providers: genuine modern spreadsheet files, legacy binary spreadsheet files, and HTML tables saved with a spreadsheet file extension — the last of these is common among exports from government statistical portals and cannot be read as a binary spreadsheet without producing a parsing error. Each series was then validated on import: date range, observation count, and missing-value count were checked against expectations before proceeding to merging.

The 35 base variables were assembled into four intermediate files, merged by country/category group before being combined into the master dataset in Chapter 2.

Table 1.2: Data dictionary: 35 base variables

ID	Variable	Country	Source
1	kz_corp_yield_m	KZ	KASE
2	ru_corp_yield_m	RU	MOEX
3–5	kz_govt_yield_{1,5,10}y_m	KZ	KASE (NSS curve parameters)
6–8	ru_govt_yield_{1,5,10}y_m	RU	MOEX (ZCYC parameters)
9	kz_tonia_m	KZ	KASE
10	kz_usdfx_m	KZ	National Bank of Kazakhstan
11	kz_inflation_yoy_m	KZ	Bureau of National Statistics
12	kz_gdp_m	KZ	taldau.stat.gov.kz
13	kz_extdebt_m	KZ	National Bank of Kazakhstan
14	kz_reserves_m	KZ	National Bank of Kazakhstan
15	kz_m2_m	KZ	National Bank of Kazakhstan
16	kz_indprod_m	KZ	taldau.stat.gov.kz
17	kz_unemp_m	KZ	taldau.stat.gov.kz
18	kz_stockidx_m	KZ	KASE
19	ru_ruonia_m	RU	CBR
20	ru_usdfx_m	RU	CBR
21	ru_inflation_yoy_m	RU	rateinflation.com
22	ru_gdp_m	RU	FRED (NGDPRNSAXDCRUQ)
23	ru_extdebt_m	RU	CBR
24	ru_reserves_m	RU	CBR
25	ru_m2_m	RU	CBR
26	ru_indprod_m	RU	Rosstat
27	ru_unemp_m	RU	Rosstat
28	ru_stockidx_m	RU	MOEX
29	gl_oil_brent_m	Global	FRED
30	gl_gold_m	Global	investing.com
31	us_yield_1y_m	Global	FRED
32	us_yield_5y_m	Global	FRED
33	us_yield_10y_m	Global	FRED
34	us_fedfunds_m	Global	FRED
35	gl_vix_m	Global	CBOE

1.5 Dependent Variables: Corporate and Government Bond Yields

Corporate bond yield indices (Kazakhstan, Russia) and government bond yields at 1, 5, and 10-year maturities (Kazakhstan, Russia) were loaded and merged on date. Kazakhstan's government yield curve parameters were sourced as decimal fractions (e.g., 0.1469) rather than percentage points; these were multiplied by 100 prior to merging so that all eight yield series share a common percentage-point scale, confirmed via a median-value check across all columns after merging. The four component series begin at different dates — Kazakhstan's corporate yield from 2000, Russia's corporate yield from 2013, Kazakhstan's government yields only from November 2019, Russia's government yields from 2003 — so an outer join was used to preserve each series' full available history rather than truncating to the shortest. The resulting file spans 314 monthly observations (July 2000 to August 2026) across 8 yield series.

1.6 Kazakhstan Domestic Fundamentals

Ten series (TONIA, the KZT/USD exchange rate, inflation, GDP, external debt, international reserves, M2 money supply, industrial production, unemployment, and the KASE stock index) were loaded and merged. Cleaning steps specific to this group: a column-naming inconsistency in the inflation series was corrected; several series (GDP, external debt, the stock index) were exported with comma-formatted numeric strings and were converted to numeric type; trailing zero values in GDP corresponding to not-yet-published periods were replaced with missing values rather than treated as genuine zero growth; inflation was converted from a decimal fraction to percentage points; and external debt, sourced at genuinely quarterly frequency, was reindexed to monthly frequency with linear interpolation between quarter-end observations (Section 1.2). The merged file spans 385 monthly observations (August 1994 to August 2026) across 10 series.

1.7 Russia Domestic Fundamentals

The corresponding ten Russian series (RUONIA, the RUB/USD exchange rate, inflation, GDP, external debt, international reserves, M2 money supply, industrial production, unemployment, and the MOEX stock index) were loaded and merged using the same procedure. A trailing whitespace character in one column name (GDP) was removed, and comma-formatted numeric strings were converted as in Section 1.6. Inflation and unemployment were converted from decimal fractions to percentage points using an idempotent, median-based check — the conversion is applied only if a series' median value indicates it is still expressed as a fraction, so the procedure is safe to re-run without risk of applying the conversion twice. The merged file spans 405 monthly observations (December 1992 to August 2026) across 10 series.

1.8 Global and Common Factors

Seven series common to both countries' models — Brent crude oil, gold, US Treasury yields at 1, 5, and 10-year maturities, the US Federal Funds Rate, and the VIX — were loaded and merged directly; no unit or format corrections were required for this group. The merged file spans 472 monthly observations (May 1987, the start of the Brent oil series, to August 2026) across 7 series.

1.9 Summary

1. All 35 base variables were sourced from primary providers and brought to monthly frequency prior to import, using the aggregation and disaggregation rules summarized in Table 1.1.

2. Four intermediate files were constructed — dependent variables (314 obs., 8 series), Kazakhstan fundamentals (385 obs., 10 series), Russia fundamentals (405 obs., 10 series), and global/common factors (472 obs., 7 series) — each merged via outer join to preserve full available history across mismatched start dates.

3. Series-specific corrections applied at this stage (unit standardization, comma-formatted numeric parsing, quarterly-to-monthly interpolation, decimal-to-percentage conversion) were necessary prerequisites for the four files to be safely merged into the single master dataset constructed in Chapter 2.

Chapter 2

Data Construction: Merging, Stationarity Testing, and Variable Construction

2.1 Data Sources and Merging

Four pre-cleaned monthly data files — dependent variables, Kazakhstan domestic fundamentals, Russia domestic fundamentals, and global/common factors — were merged on date via an outer join, preserving the full available history rather than truncating to the shortest series. An outer join is necessary here because the four source series begin at different dates (Brent oil from 1987; Russian corporate yields from 2013; Kazakhstan government yields only from 2019), and truncating to a common start would discard the majority of the historical sample unnecessarily. The resulting master dataset spans 472 monthly observations (May 1987 to August 2026) across 35 base variables. Each of the four source files, and the merged dataset itself, was checked for duplicate dates as a data-integrity control; none were found.

2.2 Stationarity Testing Methodology

Each of the 35 base variables was tested for stationarity using two complementary tests: the Augmented Dickey-Fuller (ADF) test, whose null hypothesis is that the series has a unit root, and the KPSS test, whose null hypothesis is that the series *is* stationary — the reverse null of ADF. A series was classified as stationary only when both tests agree (ADF rejects its null *and* KPSS fails to reject its null). Requiring agreement between two tests with opposite nulls is a stricter standard than relying on either test alone, since ADF has known low power against near-unit-root alternatives, while KPSS tends to over-reject stationarity in small samples.

Rather than choosing whichever transform happens to pass the tests, each variable was assigned an economically motivated default transform based on its underlying data-generating behavior (Table 2.1), and statistical testing was used to *validate* that default rather than to pick between candidates arbitrarily.

Table 2.1: Default transform by variable type

Variable type	Default transform	Rationale
Rates, yields, spreads (already in % points)	First difference	A change in percentage points is directly interpretable and is the standard unit in fixed-income and monetary economics.
Price levels, indices, monetary aggregates (near-exponential growth)	Log-difference	Approximates a % growth rate and stabilizes variance left behind by a raw first difference.

Where the preferred transform failed both tests, the alternative was tried as a fallback; where neither passed, the variable was flagged for manual resolution (Section 2.4) rather than silently forced into a transform that does not achieve stationarity.

2.3 Stationarity Results

Of the 35 variables, 3 were already stationary at the raw level, 28 achieved stationarity under their economically preferred first-order transform, and 4 — `kz_extdebt_m`, `kz_m2_m`, `ru_extdebt_m`, `ru_m2_m` — did not, and required further resolution.

2.4 Resolving Non-Stationarity in External Debt and Money Supply

For the four unresolved variables, a second log-difference was tested (to check whether the series are integrated of order 2, I(2), rather than I(1)), alongside a Zivot-Andrews unit root test on the first log-difference (which allows for one endogenously determined structural break, to check whether a single break — rather than genuine I(2) behavior — was the source of the earlier non-stationarity).

All four series achieve stationarity under a second log-difference, indicating they are integrated of order 2 rather than order 1. Two series (`kz_extdebt_m`, `kz_m2_m`) additionally show a statistically significant single structural break; two do not. Notably, **every detected break date clusters around 2007–2008**, coinciding with the Global Financial Crisis rather than the 2022 war onset that is the focus of this thesis. This indicates the four series' historical non-stationarity reflects long-run monetary and external-debt growth dynamics over a multi-decade sample, not a war-related structural shift, and second-order differencing is the appropriate resolution for use in subsequent modeling.

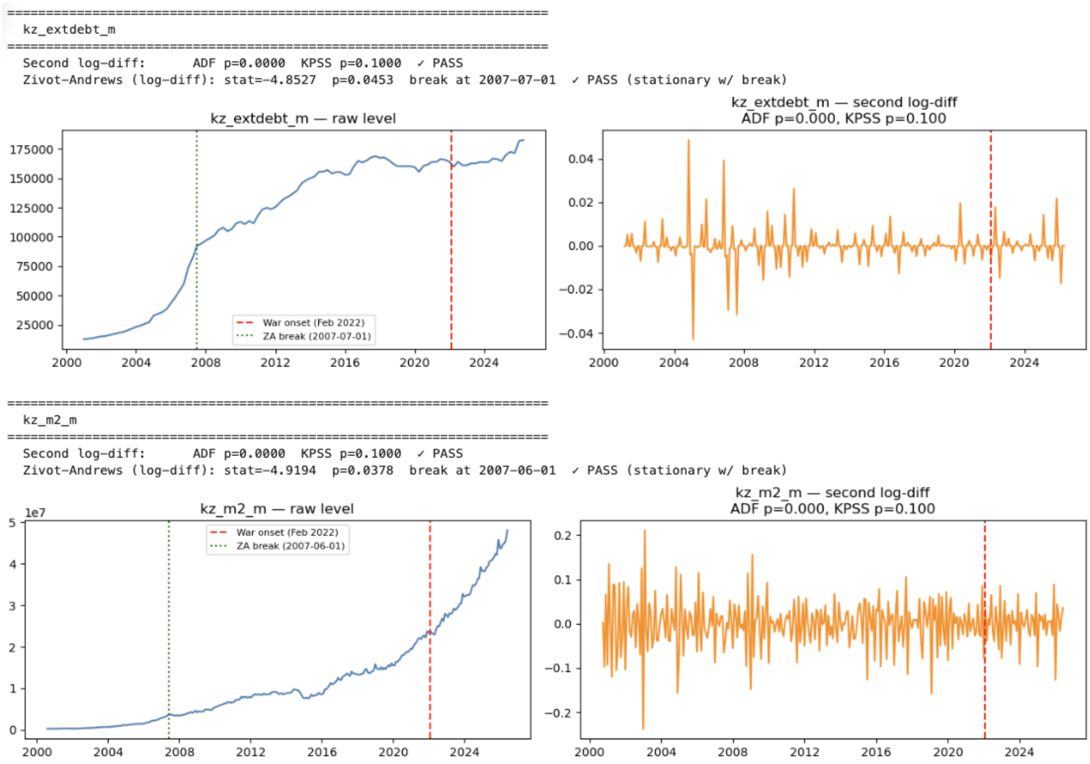

Figure 2.1: Second log-difference and Zivot-Andrews break test, Kazakhstan external debt and M2 money supply. The dashed red line marks the war onset (February 2022); the dotted green line marks the detected structural break.

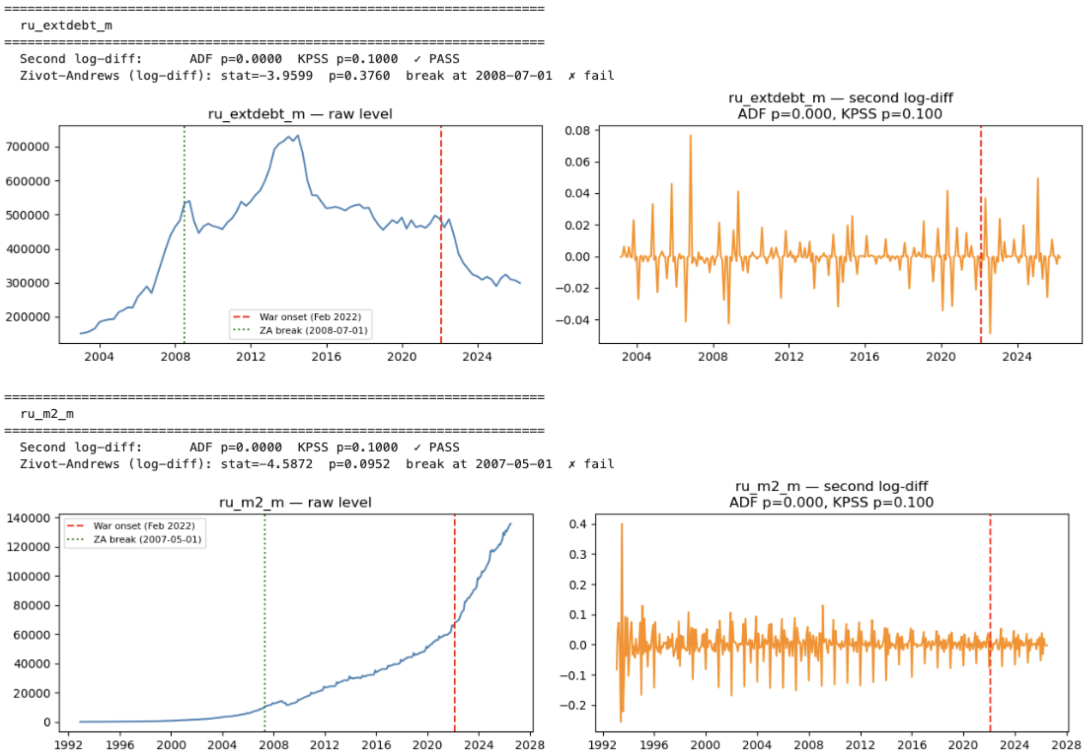

Figure 2.2: Second log-difference and Zivot-Andrews break test, Russia external debt and M2 money supply. The dashed red line marks the war onset (February 2022); the dotted green line marks the detected structural break.

2.5 Constructing the Credit-Risk Spreads

The dependent variables — corporate credit-risk spreads — are constructed as the corporate bond yield index minus the government bond yield at matched maturity, for each of the three maturities (1, 5, and 10 years) in each country:

$$\text{spread}_{i,m,t} = \text{corp\_yield}_{i,t} - \text{govt\_yield}_{i,m,t}, \qquad i \in \{\text{KZ}, \text{RU}\},\ m \in \{1, 5, 10\}. \tag{2.1}$$

This construction isolates the credit-risk premium by netting out the common risk-free/term-structure component shared by both bonds.

2.6 Constructing the Fiscal Ratios: Unit and Flow/Stock Corrections

Two fiscal ratios — external debt to GDP and reserves to M2 — were constructed from the underlying level series. Two corrections were required before these ratios were economically meaningful.

Currency units. External debt and reserves are denominated in USD, while M2 and GDP are denominated in local currency (tenge, ruble). M2 was converted to USD using the corresponding exchange rate series before forming the reserves-to-M2 ratio; GDP was converted to USD on the same basis before forming the external-debt-to-GDP ratio.

Flow vs. stock. External debt is a *stock* variable, while GDP as sourced is a *monthly flow* (disaggregated from quarterly data). Dividing a debt stock directly by a monthly GDP flow overstates the conventional debt-to-GDP ratio by roughly a factor of 12. This was corrected by annualizing GDP via a trailing 12-month rolling sum before forming the ratio:

$$\mathrm{ExtDebt/GDP}_t = \frac{\mathrm{ExtDebt}_t}{\sum_{i=0}^{11} \mathrm{GDP}_{t-i}}. \tag{2.2}$$

This introduces 11 months of missing values at the start of each GDP series, where a full trailing window is not yet available — treated as a genuine data limitation rather than filled or estimated. No corresponding adjustment was required for reserves-to-M2, since both underlying series are stocks.

The corrected external-debt-to-GDP ratios (Kazakhstan ≈85% of GDP, Russia ≈23%) are consistent with each country's known external indebtedness profile, confirming the correction was necessary and economically sensible.

2.7 Diagnosing Negative Credit Spreads

A negative spread (corporate yield below government yield) is economically unusual and warrants investigation rather than automatic adjustment.

Two distinct patterns emerge, interpreted differently. **Kazakhstan (1yr, 5yr):** negative observations cluster almost entirely in the period immediately following February 2022, extending through 2023, consistent with the National Bank of Kazakhstan sharply raising short-term policy rates in response to war-driven

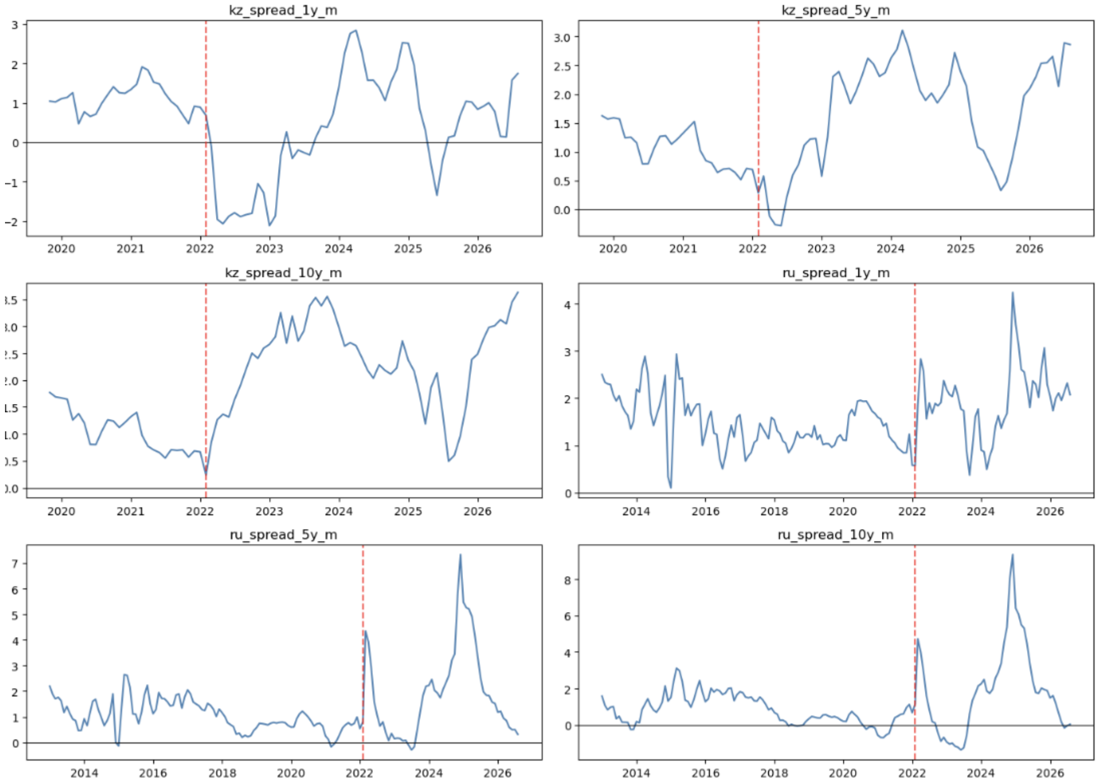

Figure 2.3: All six constructed credit-risk spreads, with the war onset (dashed red line) and zero (solid black line) marked.

capital outflows and inflation, with short-maturity government yields repricing faster than the corporate bond index. This is treated as a substantive finding relevant to the research question, not a data artifact, and is retained without adjustment. **Russia (10yr):** negative observations recur intermittently across the full sample with no concentration around 2022, and are attributed to the construction of the underlying government yield series — sourced from MOEX's published zero-coupon yield curve (ZCYC) *parameters*, a fitted curve rather than directly observed long-maturity transaction yields, which can behave irregularly at the long end during thin trading. This is documented as a data-construction caveat rather than an economically meaningful result. In neither case were negative values removed, winsorized, or floored at zero, since doing so would discard genuine information.

2.8 War Dummy and Volatility Construction

A binary war indicator (`war_dummy` = 1 from February 2022 onward, 0 before) was constructed directly from the calendar date, yielding 417 pre-war and 55 post-war monthly observations.

Stock market volatility was constructed as the rolling 3-month standard deviation of monthly log returns on each country's stock index:

$$\mathrm{vol}_{i,t} = \mathrm{sd}\left(\Delta \log(\mathrm{stockidx}_{i,\tau})\right)_{\tau=t-2}^{t}. \tag{2.3}$$

Since the underlying stock index data is monthly rather than daily, this is a rolling window of monthly returns rather than realized volatility computed from intraday or daily data; the latter would be a materially better proxy if higher-frequency data were available. The full time series of this volatility measure, and its interpretation relative to the war onset, is presented in Chapter 3.

2.9 Final Dataset Assembly

The base variables (Section 2.3), the resolved I(2) variables (Section 2.4), the constructed spreads (Section 2.5), the fiscal ratios (Section 2.6), the war dummy, and the volatility measures (Section 2.8) were assembled into a single workbook with three sheets: *Levels (Raw)*, containing every constructed variable traceable back to its source inputs; *Transformed (Stationary)*, containing the stationarity-tested series used in subsequent estimation; and a *Codebook*, recording the transform applied and final test statistics for every variable.

As a final verification step, the transform recorded for each of the six credit-risk spreads was confirmed directly against the saved codebook: Kazakhstan's three spreads all resolved to `diff` (confirming I(1) behavior), while Russia's three spreads resolved to `none needed` (confirming I(0) behavior at the raw level). This mixed order of integration across the two countries' dependent variables is the direct motivation for the ARDL bounds-testing framework used in Chapter 4.

2.10 Summary

1. Four source files were merged into a single 472-observation, 35-variable monthly dataset spanning May 1987 to August 2026, with no duplicate dates in any source file or in the merged result.

2. Stationarity was established for all 35 base variables using an economically motivated default transform, validated by requiring agreement between ADF and KPSS; four variables required a second log-difference, resolved as I(2) rather than I(1), with structural breaks (where detected) clustering around the 2007–2008 financial crisis rather than the 2022 war.

3. Credit-risk spreads were constructed as corporate minus government yield at matched maturity; fiscal ratios were constructed with explicit currency-unit and flow/stock corrections, without which the resulting ratios would have been economically implausible.

4. Negative spread observations were diagnosed rather than adjusted: Kazakhstan's pattern is war-related and substantively informative; Russia's is a pre-existing artifact of the underlying yield-curve construction.

5. The final dataset confirms Kazakhstan's spreads are I(1) and Russia's are I(0) — the mixed order of integration that motivates the ARDL bounds-testing approach used in Chapter 4, and that underlies the exploratory analysis of Chapter 3.

Table 2.2: Stationarity test results and applied transforms, 35 base variables

Variable	Transform applied	ADF p (final)	KPSS p (final)
kz_corp_yield_m	diff	0.000	0.100
ru_corp_yield_m	diff	0.000	0.100
kz_govt_yield_1y_m	diff	0.000	0.100
kz_govt_yield_5y_m	diff	0.000	0.100
kz_govt_yield_10y_m	diff	0.001	0.100
ru_govt_yield_1y_m	diff	0.000	0.100
ru_govt_yield_5y_m	diff	0.000	0.100
ru_govt_yield_10y_m	diff	0.000	0.100
kz_tonia_m	diff	0.000	0.100
kz_usdfx_m	logdiff	0.000	0.100
kz_inflation_yoy_m	diff	0.002	0.100
kz_gdp_m	logdiff	0.000	0.100
kz_extdebt_m	second logdiff*	0.000	0.100
kz_reserves_m	logdiff	0.000	0.100
kz_m2_m	second logdiff*	0.000	0.100
kz_indprod_m	logdiff	0.000	0.100
kz_unemp_m	diff	0.000	0.100
kz_stockidx_m	logdiff	0.000	0.100
ru_ruonia_m	diff	0.000	0.100
ru_usdfx_m	logdiff	0.000	0.100
ru_inflation_yoy_m	diff	0.000	0.100
ru_gdp_m	logdiff	0.004	0.100
ru_extdebt_m	second logdiff*	0.000	0.100
ru_reserves_m	diff	0.000	0.100
ru_m2_m	second logdiff*	0.000	0.100
ru_indprod_m	logdiff	0.003	0.100
ru_unemp_m	diff	0.000	0.100
ru_stockidx_m	logdiff	0.000	0.100
gl_oil_brent_m	logdiff	0.000	0.100
gl_gold_m	logdiff	0.000	0.053
us_yield_1y_m	none needed	0.004	0.076
us_yield_5y_m	diff	0.000	0.081
us_yield_10y_m	diff	0.000	0.094
us_fedfunds_m	none needed	0.004	0.086
gl_vix_m	none needed	0.000	0.100

**Resolved via second-order log-differencing after the preferred first-order transform failed; see Section 2.4.*

Table 2.3: Resolution of the four non-stationary variables

Variable	2nd log-diff (ADF/KPSS p)	Zivot-Andrews p-value	Detected break date	Resolution
kz_extdebt_m	0.000 / 0.100	0.045	2007-07	2nd log-diff
kz_m2_m	0.000 / 0.100	0.038	2007-06	2nd log-diff
ru_extdebt_m	0.000 / 0.100	0.376	2008-07	2nd log-diff
ru_m2_m	0.000 / 0.100	0.095	2007-05	2nd log-diff

Table 2.4: Constructed fiscal ratios: summary statistics

Variable	n	Mean	Std. Dev.	Min	Max
kz_extdebt_gdp_ratio	136	0.851	0.182	0.594	1.245
ru_extdebt_gdp_ratio	124	0.233	0.034	0.165	0.290
kz_reserves_m2_ratio	311	0.757	0.200	0.412	1.283
ru_reserves_m2_ratio	199	0.644	0.112	0.401	0.922

Table 2.5: Negative spread observations by series

Series	Negative observations
kz_spread_1y_m	20
kz_spread_5y_m	3
kz_spread_10y_m	0
ru_spread_1y_m	0
ru_spread_5y_m	6
ru_spread_10y_m	30

Chapter 3

Exploratory Data Analysis: Pre- and Post-War Patterns

3.1 Data Overview

This chapter conducts exploratory data analysis on the final merged and stationarity-tested dataset produced in Chapter 2 — summary statistics, correlation structure, and time series visualization, split by the pre-war and post-war periods — to establish a preliminary empirical picture of the research question before formal estimation in Chapters 4–6 (ARDL, Markov-switching, VAR/IRF).

The dataset consists of two parallel sheets: a *Levels* sheet (472 monthly observations, 56 variables, spanning May 1987 to August 2026) and a *Transformed* sheet (471 observations, 48 variables, each series stationarity-tested and differenced or left at level as required). The eight-variable gap between the two sheets is expected, not an error: it corresponds to intermediate helper variables constructed solely to build the fiscal ratios and volatility measures (`kz_m2_usd`, `ru_m2_usd`, `kz_gdp_usd`, `ru_gdp_usd`, `kz_gdp_annual_usd`, `ru_gdp_annual_usd`, `kz_stockidx_ret`, `ru_stockidx_ret`). These are retained in the Levels sheet for full traceability but were never intended as modeling variables in their own right, and correctly have no stationarity-transformed counterpart. The single-row difference between the sheets (472 vs. 471) is the expected consequence of first-differencing, which necessarily drops the first observation of the sample.

3.2 Pre- and Post-War Summary Statistics

Kazakhstan's credit-risk spread series begin only in November 2019, giving a pre-war sample of 27 observations (through January 2022), compared to 109 for Russia (spread series beginning January 2013). All comparisons involving Kazakhstan's pre-war baseline throughout this thesis should be read with this limited sample in mind.

Table 3.1: Pre-war summary statistics (before February 2022)

Variable	n	Mean	Std. Dev.	Min	Max
`kz_spread_1y_m`	27	1.112	0.364	0.473	1.912
`kz_spread_5y_m`	27	1.078	0.348	0.516	1.625
`kz_spread_10y_m`	27	1.056	0.380	0.552	1.770
`ru_spread_1y_m`	109	1.464	0.537	0.102	2.937
`ru_spread_5y_m`	109	1.047	0.592	−0.174	2.649
`ru_spread_10y_m`	109	0.835	0.804	−0.691	3.117
`kz_vol_m`	256	0.048	0.046	0.002	0.314
`ru_vol_m`	262	0.050	0.037	0.003	0.244

Table 3.2: Post-war summary statistics (February 2022 onward)

Variable	n	Mean	Std. Dev.	Min	Max
`kz_spread_1y_m`	55	0.327	1.389	−2.111	2.839
`kz_spread_5y_m`	55	1.693	0.921	−0.284	3.108
`kz_spread_10y_m`	55	2.328	0.830	0.235	3.628
`ru_spread_1y_m`	55	1.928	0.735	0.371	4.241
`ru_spread_5y_m`	55	1.922	1.765	−0.291	7.335
`ru_spread_10y_m`	55	1.913	2.428	−1.360	9.346
`kz_vol_m`	55	0.026	0.016	0.003	0.083
`ru_vol_m`	55	0.046	0.032	0.005	0.146

Key findings. Russian credit spreads rose substantially post-war across all three maturities, with mean and standard deviation both increasing sharply (10yr mean: 0.84 → 1.91; std. dev.: 0.80 → 2.43) — consistent with H2's expectation of a larger, persistent Russian break. Kazakhstan's spreads show a maturity-dependent pattern instead: the 1yr spread's mean *fell* post-war (driven by a documented negative-spread episode in 2022–23, discussed further in Section 3.4), while the 5yr and 10yr spreads rose — indicating the war's effect on Kazakhstan's credit market is not uniform across the term structure, a finding carried forward into the ARDL specification (Chapter 4).

Stock index volatility fell (Kazakhstan) or held roughly flat (Russia) on average post-war, which is counterintuitive for a shock period and is addressed directly in Section 3.7 once the full volatility time series is examined.

3.3 Correlation Structure and the Transmission Mechanism

To examine whether the relationship between each country's credit spread and its macro-financial determinants changed after the invasion, correlation matrices were computed separately for the pre-war and post-war periods, using a common driver set: domestic interbank liquidity rates (TONIA, RUONIA), global risk factors (Brent oil, VIX, the US 10-year Treasury yield), and local stock market volatility.

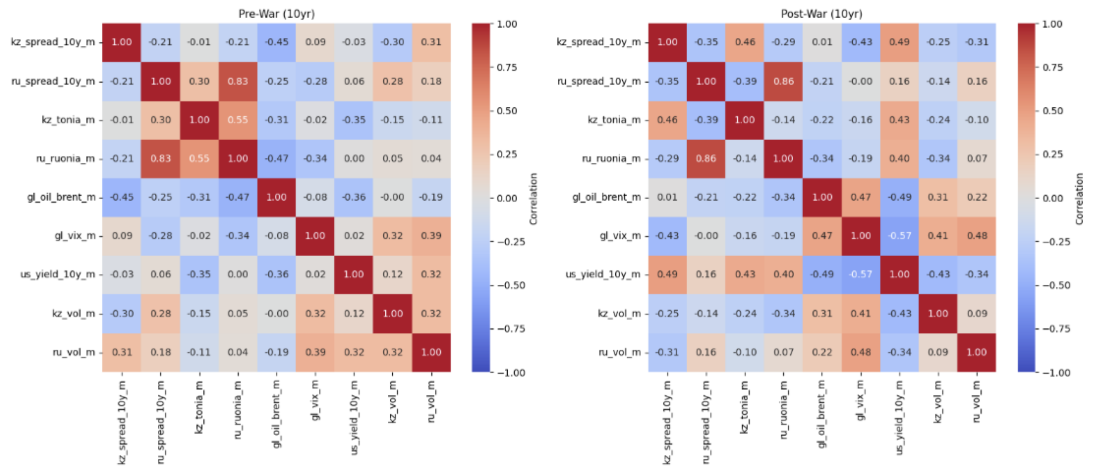

Figure 3.1: 10-year credit spread correlation matrix, pre-war vs. post-war. Complete-case observations: 27 pre-war, 55 post-war. Pre-war period: 1987-05-01 to 2022-01-01; post-war period: 2022-02-01 to 2026-08-01.

10-year maturity (main result). Kazakhstan's 10yr spread shows a striking oil decoupling: correlation with Brent crude fell from -0.45 pre-war to ~ 0.01 post-war, consistent with a loss of the historical (oil-exporter) relationship between fiscal strength and credit risk. Russia's spread-oil relationship, by contrast, was stable ($-0.25 \rightarrow -0.21$), suggesting the oil-spread decoupling is specific to Kazakhstan, not a regional or global pattern affecting both countries equally. Kazakhstan's spread-TONIA (domestic liquidity) correlation strengthened substantially ($-0.01 \to 0.46$), suggesting a shift toward domestic liquidity conditions as a driver, coinciding with the oil decoupling above. Russia's spread-RUONIA relationship remained strong and largely unchanged ($0.83 \to 0.86$) both pre- and post-war. The KZ–RU spread cross-correlation became *more* negative ($-0.21 \to -0.35$), not closer to zero — a distinct finding from "transmission mechanism divergence" that should be interpreted carefully, since a stronger inverse relationship is not the same claim as a weaker one.

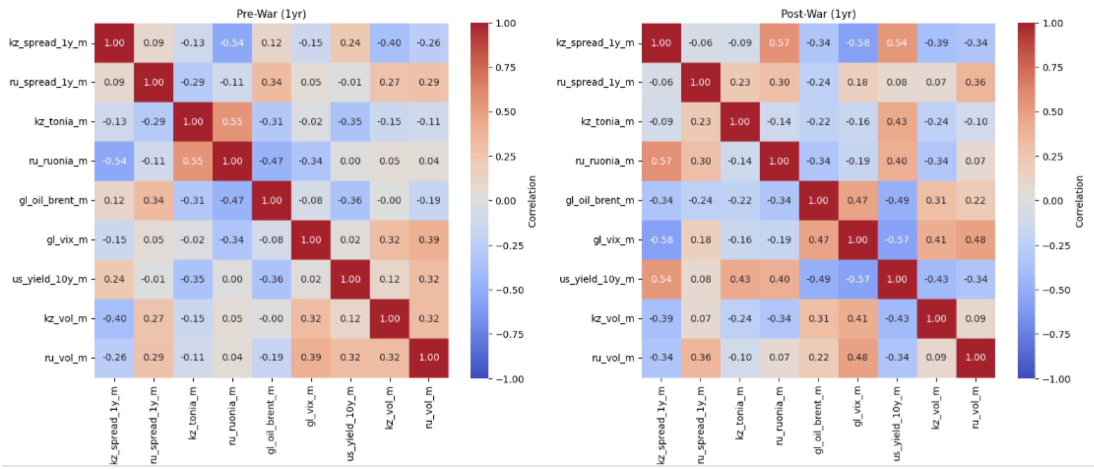

Figure 3.2: 1-year credit spread correlation matrix, pre-war vs. post-war — robustness check using the same driver set as Figure 3.1.

Maturity dependence (Figures 3.2 and 3.3). The 10yr oil-decoupling and liquidity-strengthening pattern does *not* hold uniformly across maturities — it strengthens with maturity rather than applying as a uniform shift. The oil correlation at 1yr moved *more negative* post-war ($0.12 \to -0.34$, the opposite direction from decoupling); the 5yr correlation showed partial decoupling ($-0.41 \to -0.21$); only the 10yr correlation showed full decoupling ($-0.45 \to 0.01$). The oil-decoupling finding should therefore be described as a longer-horizon phenomenon, not a blanket feature of Kazakhstan's credit spread. The domestic liquidity (TONIA) correlation shows the same pattern: flat at 1yr ($-0.13 \to -0.09$), but strengthening at 5yr and 10yr ($-0.04 \to 0.32$; $-0.01 \to 0.46$).

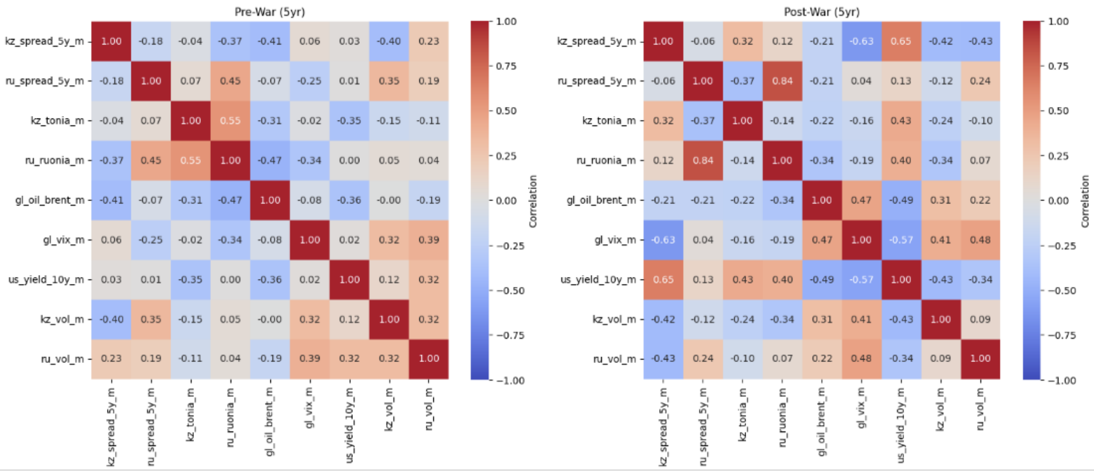

Figure 3.3: 5-year credit spread correlation matrix, pre-war vs. post-war — robustness check using the same driver set as Figure 3.1.

One notable outlier deserves mention: `kz_spread_1y_m` versus `ru_ruonia_m` flipped sign entirely (-0.54 pre-war $\rightarrow +0.57$ post-war), the largest single swing across all three matrices. Given the 1yr spread's documented negative-value episode in 2022–23 (Section 3.4), this result should be interpreted cautiously — it may partly reflect that same anomalous period rather than a stable new relationship.

Implication. The transmission-mechanism shift this analysis is designed to detect appears to be a maturity-dependent, long-end phenomenon in Kazakhstan's credit market, not a uniform shift across the curve. This is itself a substantive finding and is stated as such throughout this thesis, rather than presenting only the 10yr result as if it generalized to the full term structure.

3.4 Time Series of Credit Spreads

Figure 3.4 shows Russia's spread exhibiting the classic shock signature at war onset: a sharp spike immediately following February 2022 across all three maturities, receding within months. Kazakhstan's initial reaction is markedly different

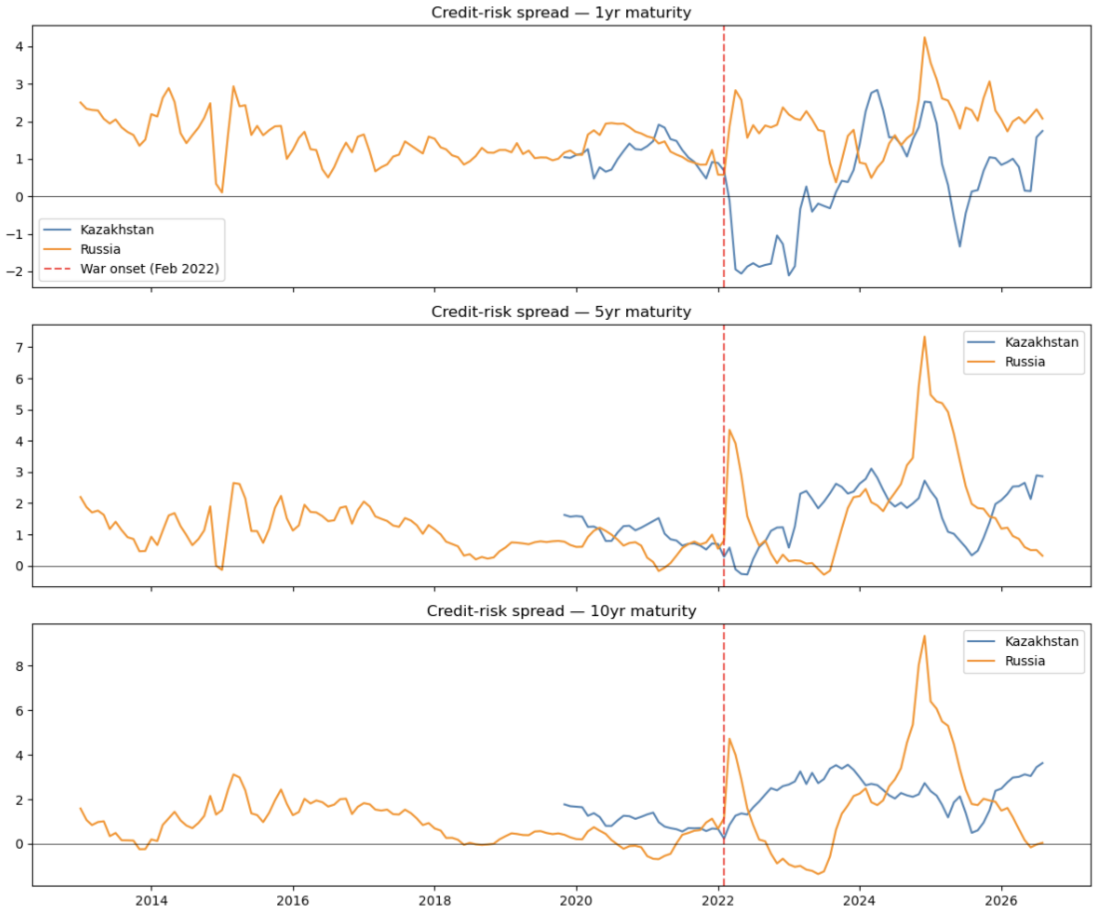

Figure 3.4: Credit-risk spreads, Kazakhstan vs. Russia, all three maturities, 1987–2026 (Kazakhstan series begin November 2019). The dashed red line marks the war onset (February 2022).

— a temporary *drop* into negative territory at the 1yr and 5yr maturities immediately post-war, followed by a slower, gradual rise through 2023–2026 rather than a sharp shock.

A second, larger Russian spread spike occurs around 2024–2025 (5yr peaks at ∼7.3, 10yr peaks at ∼9.3), exceeding the magnitude of the initial February 2022 reaction. This is a substantial finding not addressed by a single-break (February 2022) framing and is investigated further in Section 3.6; it is a natural candidate for a second regime the Markov-switching model (Chapter 5) might be expected to detect, rather than assuming a single clean regime change. Kazakhstan and Russia's spreads also show visually apparent anti-comovement at several points post-war (Russia high while Kazakhstan is low, and vice versa), consistent with the increasingly negative KZ–RU cross-correlation found in Section 3.3.

3.5 Rolling Correlation Between Kazakhstan and Russia

Both window lengths are plotted together to compare directly. The 12-month window reveals genuine cyclical structure that the 24-month window smooths away: rolling correlation between the two countries' spreads swings repeatedly between strongly negative (~ -0.8 to -0.9) and strongly positive ($\sim +0.65$ to $+0.85$) roughly every 12–18 months post-war, rather than settling into one stable post-break state. The 10yr panel shows the clearest cyclical pattern, with swings recurring around early 2023, late 2023, early 2025, and late 2025/2026. The strongest positive correlation spike in the entire post-war sample occurs in the 10yr panel in late 2025 (∼0.83, 12-month window), coinciding with the second large Russian spread spike identified in Section 3.4 — reinforcing that this 2024–25 episode drove Kazakhstan's and Russia's spreads into brief alignment rather than further divergence.

This finding materially changes the paper's framing: the war did not create one static "before vs. after" regime for KZ–RU comovement, but rather a more volatile, cyclically-shifting relationship that briefly reconverges during shared

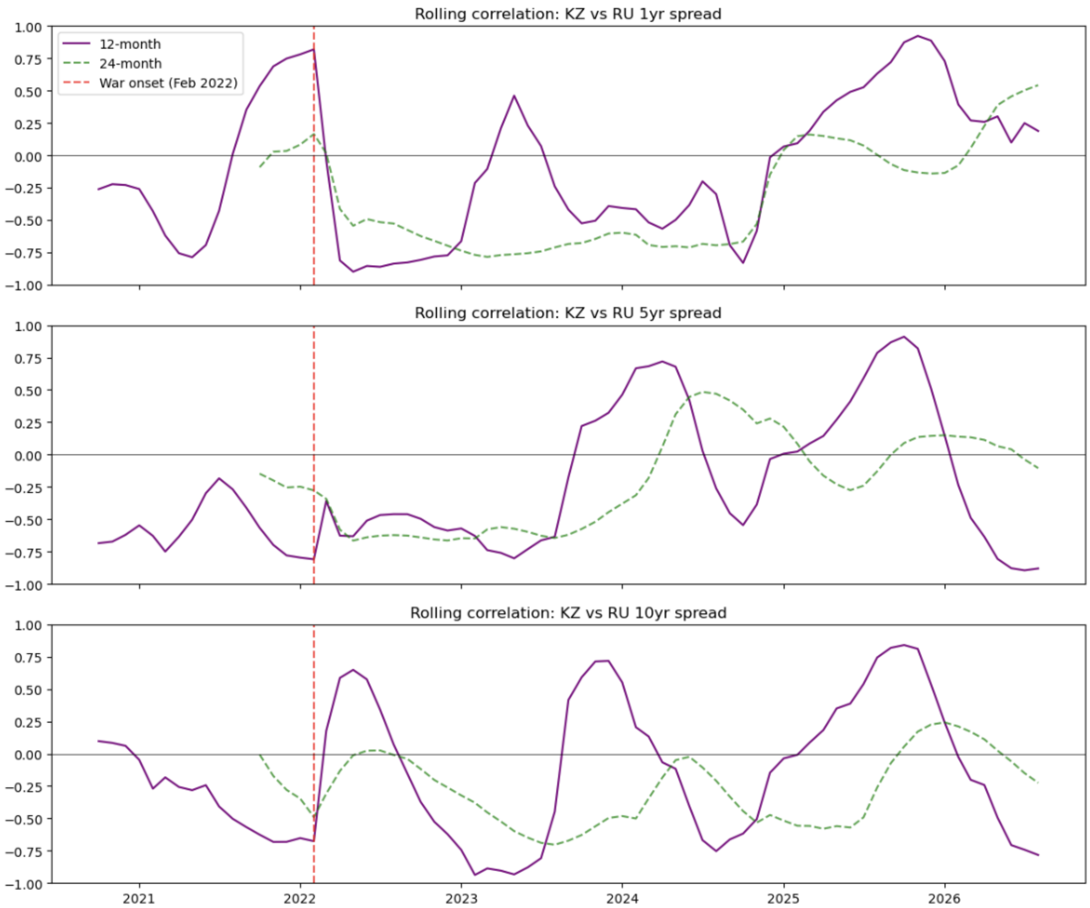

Figure 3.5: 12-month (solid) and 24-month (dashed) rolling correlation between Kazakhstan's and Russia's credit spreads, all three maturities. The dashed red line marks the war onset (February 2022).

shock episodes. This has direct implications for the Markov-switching specification (Chapter 5), which should allow for more than two regimes, or investigate whether regime transitions are more frequent than a single February-2022 break would imply. The 24-month line is retained alongside the 12-month line specifically to demonstrate how a wider window can obscure the shorter-cycle dynamics the 12-month line reveals.

3.6 The Second Shock: Russia's Late-2024 Fiscal Strain

The second, larger spike in Russia's credit spread identified in Sections 3.4 and 3.5 corresponds to a documented deterioration in Russian government bond market conditions: Russian short-term government bond yields reached an all-time high of 22.78% in December 2024, driven by the Bank of Russia's aggressive monetary tightening in response to inflation from sustained wartime government spending.[1]

This is best understood as a **structural fiscal story rather than a discrete sanctions event**: with international debt markets closed by sanctions, the Russian government has relied increasingly on expensive domestic borrowing, producing a sustained deterioration — heavy OFZ issuance, a widening budget deficit driven by defense spending roughly 40% above initial targets, and debt-service costs projected to consume at least 15% of GDP over the next decade. This is consistent with a persistent, compounding effect of the war on Russian credit conditions (supporting H2), rather than the initial 2022 shock alone.

Implication. The war's effect on Russian credit-risk pricing should be framed as an ongoing structural strain — driven by fiscal and monetary dynamics tied to sustained wartime spending — rather than a single February 2022 shock that fully resolved. This has direct implications for the Markov-switching model (Chapter 5), which may detect a second, distinct regime shift around late 2024, and for how this thesis's conclusion frames "persistence" in H2.

[1]Source: Trading Economics, Russia 10-Year Government Bond Yield.

3.7 Equity Market Volatility

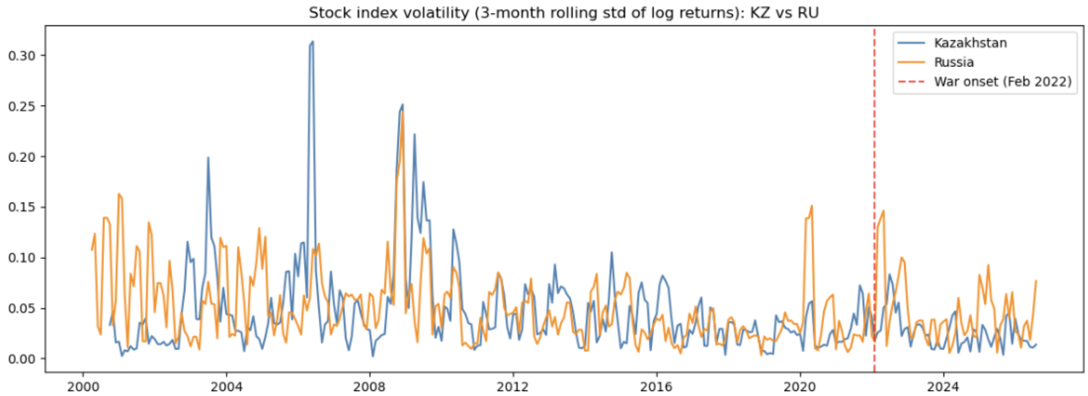

Figure 3.6: Stock index volatility (3-month rolling standard deviation of log returns), Kazakhstan vs. Russia, 2000–2026. The dashed red line marks the war onset (February 2022).

Figure 3.6 resolves the counterintuitive summary-statistic finding of Section 3.2 (average volatility appeared to fall post-war): the full 26-year series shows the February 2022 war-onset volatility bump is real but modest, and substantially smaller than several earlier episodes already present in the pre-war sample — most notably Kazakhstan's own 2006 spike (∼0.31, the largest value in the entire series), and the 2008 global financial crisis (both countries reaching ∼0.20–0.25). A 2020 COVID-era spike in Russia (∼0.15) also exceeds the war-onset reaction.

Implication. Unlike the credit-spread results, which show clear and substantial war-specific effects, stock market volatility does not single out the war as an unusual event relative to Kazakhstan's and Russia's own historical volatility experience. This is stated explicitly rather than overclaimed: it sharpens this thesis's argument by clarifying that the war's detectable financial-market effect operates primarily through the credit-spread channel examined here, not through broad equity market volatility.

3.8 Distributional Properties and Order of Integration

Labeling clarification. The six panels in Figure 3.7 are not all the same kind of quantity. Kazakhstan's spreads required first-differencing to achieve station-

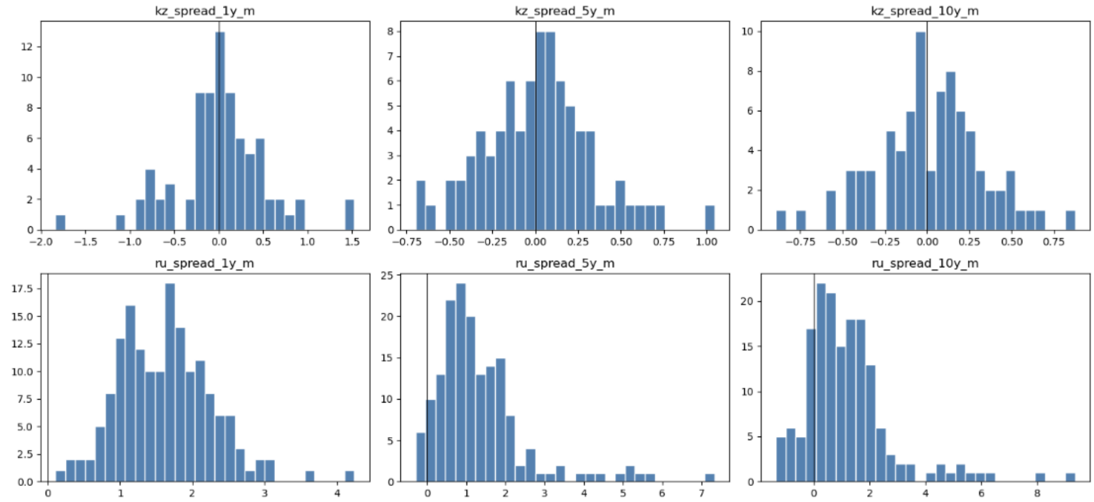

Figure 3.7: Distribution of the transformed dependent variables (stationary series used in subsequent estimation).

Table 3.3: Transformed dependent variables: distribution statistics

Variable	Mean	Std. Dev.	Min	Max
kz_spread_1y_m	0.009	0.515	−1.841	1.525
kz_spread_5y_m	0.015	0.320	−0.697	1.047
kz_spread_10y_m	0.023	0.319	−0.894	0.874
ru_spread_1y_m	1.620	0.647	0.102	4.241
ru_spread_5y_m	1.341	1.198	−0.291	7.335
ru_spread_10y_m	1.197	1.625	−1.360	9.346

arity, so their histograms show *month-to-month changes* (correctly centered near zero, means ∼0.01–0.02). Russia's spreads passed stationarity testing *at the raw level*, so their histograms show *levels*, not differences (means ∼1.2–1.6, matching the raw spread values themselves). Directly comparing a differenced series against a level series side by side would otherwise be misleading, and this distinction is maintained consistently throughout the thesis wherever these series are referenced.

Distribution shape. Kazakhstan's panels (differences) are reasonably symmetric, consistent with well-behaved stationary series suitable for the ARDL specification. Russia's panels (levels) are right-skewed, with long tails driven by the 2022 war-onset shock and the 2024–25 fiscal-strain episode (max values of 7.3 and 9.3 in the 5yr/10yr panels correspond directly to those events).

Modeling implication. Kazakhstan's spreads are integrated of order 1, I(1), while Russia's are integrated of order 0, I(0) — a mixed order of integration across the two countries' dependent variables. Rather than a problem to reconcile, this is precisely the setting ARDL bounds testing (Pesaran, Shin & Smith, 2001) is designed for: it accommodates a mix of I(0) and I(1) regressors without requiring all series to share the same order of integration, provided none are I(2) — already confirmed and resolved in Chapter 2, where four I(2) variables were resolved via second-order differencing before inclusion in the final dataset. This directly supports ARDL as the appropriate modeling choice for Chapter 4, rather than requiring an alternative unified transformation.

3.9 Summary and Implications for the Modeling Chapters

This section synthesizes the exploratory findings of Sections 3.1–3.8 into a single reference point ahead of formal estimation.

Sample coverage. Kazakhstan's credit-risk spread series begin only in November 2019, giving a thin pre-war baseline of 27 observations, versus 109 for Russia. All comparisons involving Kazakhstan's pre-war behavior should be read with this limited sample in mind.

Order of integration. Kazakhstan's spreads are I(1) (stationary only after first-differencing); Russia's spreads are I(0) (stationary at the raw level). This mixed order of integration across countries is directly compatible with, and indeed the motivating rationale for, the ARDL bounds-testing framework used in Chapter 4.

Pre-war vs. post-war levels and volatility. Russian credit spreads rose substantially post-war across all three maturities, with mean and standard deviation both increasing sharply (10yr mean: 0.84 $\rightarrow$ 1.91; std. dev.: 0.80 $\rightarrow$ 2.43). Kazakhstan's spreads show a maturity-dependent pattern: the 1yr spread's mean fell post-war (driven by a documented negative-spread episode in 2022–23), while 5yr and 10yr spreads rose — indicating the war's effect on Kazakhstan's credit market is not uniform across the term structure.

Correlation structure and transmission mechanism. Kazakhstan's 10yr spread shows a near-complete decoupling from Brent oil prices post-war ($-0.45 \rightarrow 0.01$), alongside a strengthening relationship with domestic liquidity conditions (TONIA: $-0.01 \rightarrow 0.46$). This pattern strengthens with maturity (weak at 1yr, partial at 5yr, complete at 10yr) rather than applying uniformly across the curve. Russia's spread-oil and spread-RUONIA relationships remained comparatively stable across both periods.

Cross-country comovement. Rolling correlation between Kazakhstan and Russia's spreads dropped sharply and durably negative immediately following the war onset, but did not settle into a new stable state — it oscillates repeatedly between strongly negative and strongly positive values on a roughly 12–18 month cycle throughout the post-war period, with the strongest positive spike ($\sim$0.83 at 10yr) occurring in late 2025.

A second shock episode. Time series and rolling-correlation evidence both point to a second, distinct shock to Russian credit markets around late 2024, larger in magnitude than the initial February 2022 reaction. This coincides with a documented deterioration in Russian government bond markets (short-term yields reaching an all-time high of 22.78% in December 2024), driven by aggressive monetary tightening in response to inflation from sustained wartime fiscal spending. This is interpreted as a structural, compounding fiscal strain rather than a discrete sanctions event, and suggests the Markov-switching model (Chapter 5) should be evaluated for more than one regime transition, rather than assuming a single clean break at February 2022.

Equity market volatility. Unlike the credit-spread channel, Kazakhstan's and Russia's stock index volatility does not single out the war as an unusual event relative to each country's own historical volatility experience — the 2006 (Kazakhstan) and 2008 (both countries) episodes produced substantially larger volatility spikes than the 2022 war onset. This sharpens rather than weakens this thesis's argument: the war's detectable financial-market effect operates primarily through the credit-spread channel examined here, not through broad equity volatility.

Implications for the modeling chapters.

1. **ARDL (Chapter 4):** should exploit the confirmed mixed I(0)/I(1) structure directly; expect maturity-dependent results given the observed term-structure heterogeneity in Kazakhstan.

2. **Markov-switching (Chapter 5):** should not be constrained to detect only one regime shift; the evidence here supports investigating a second transition around late 2024.

3. **VAR/IRF (Chapter 6):** the oil-decoupling finding, strongest at the 10yr maturity, is a natural focus for impulse-response analysis comparing pre- and post-war transmission of oil shocks to each country's spread.

Chapter 4

ARDL Bounds Testing: Pre-War Relationships and Long-Run Dynamics

4.1 Motivation and Methodology

This chapter estimates ARDL bounds-testing models for Kazakhstan's and Russia's corporate credit-risk spreads against domestic liquidity and global macro-financial determinants, testing for a long-run (cointegrating) relationship and estimating the corresponding error-correction model (ECM). This directly addresses H1 (pre-war similarity in spread determinants) and establishes the baseline against which the Markov-switching (Chapter 5) and VAR/IRF (Chapter 6) analyses test for post-war structural change.

Why ARDL bounds testing fits this data. As established in the stationarity analysis (Chapter 3), Kazakhstan's credit spreads are integrated of order 1, I(1), while Russia's are integrated of order 0, I(0) — a mixed order of integration across the two countries' dependent variables. The ARDL bounds-testing approach (Pesaran, Shin & Smith, 2001) is specifically designed for this setting: it accommodates a mix of I(0) and I(1) regressors without requiring a common order of integration across all series, provided none are I(2) (already confirmed and resolved in Chapter 2). This avoids the need to pre-test for cointegration using methods such as Engle-Granger or Johansen, both of which require all series to share the same order of integration — a requirement this dataset does not satisfy.

Each model is estimated in its Unrestricted Error Correction (UECM) representation, from which the bounds test is computed. Under Case 3 (unrestricted constant, no trend), the test compares an F-statistic against lower (I(0)) and upper (I(1)) critical value bounds: exceeding the upper bound supports cointegration; falling below the lower bound supports no cointegration; falling between the two bounds is inconclusive.

4.2 Variable Specification

Dependent variables. Credit-risk spreads (corporate yield minus government yield) at three maturities per country: `kz_spread_1y_m`, `kz_spread_5y_m`, `kz_spread_10y_m`, and the Russian equivalents.

Regressor sets. Three specifications were tested, escalating in complexity, as summarized in Table 4.1.

Table 4.1: ARDL specifications tested

Specification	Kazakhstan regressors	Russia regressors
A (minimal)	TONIA, Brent oil, VIX	RUONIA, Brent oil, VIX
B (mixed)	TONIA, Brent oil, VIX (unchanged)	Full set*
C (full)	Full set*	Full set* (identical to B)

**Full set: liquidity rate, oil, VIX, inflation, GDP, external debt/GDP, reserves/M2, volatility.*

All models enforce a minimum lag order of 1 for every regressor (a requirement of the UECM bounds-testing procedure) and use `lags=1` for the dependent variable's own autoregressive term.

Sample splits. Two sample windows were used: (i) pre-war only (before February 2022), and (ii) the full sample with a binary war dummy (`war_dummy` = 1 from February 2022 onward) included as an additional regressor.

A binding data constraint. Kazakhstan's government bond yield series begins only in late 2019, giving a pre-war sample of just 27 observations, compared to 109 for Russia (whose spread series begins 2013). Even the minimal specification (4 regressors including the maturity-matched US Treasury yield, later dropped — see Section 4.3) requires roughly 8 parameters once lag terms are counted; the full specification requires approximately 18. Against 27 observations, the fuller specifications approach or exceed the classical rule of thumb of ∼10 observations per estimated parameter. This constraint shapes the interpretation of every Kazakhstan result in this chapter and is not a peripheral caveat.

4.3 Specification A: Minimal Model, Pre-War

Table 4.2: Specification A results, pre-war sample

Country	Maturity	n	EC term (p)	Bounds F-stat	Verdict
KZ	1yr	27	−0.072 (0.568)	2.17	No cointegration
KZ	5yr	27	−0.265 (0.068)	1.11	No cointegration
KZ	10yr	27	−0.302 (0.024)	1.99	No cointegration
RU	1yr	109	−0.318 (<0.001)	6.33	Cointegration
RU	5yr	109	−0.328 (<0.001)	8.78	Cointegration
RU	10yr	109	−0.328 (<0.001)	7.63	Cointegration

Note: the US Treasury yield was dropped from the Kazakhstan 1yr model by automatic lag-order selection in initial testing, plausibly reflecting collinearity with VIX/oil given the small sample; it was excluded from all subsequent Specification A runs across maturities for consistency.

Findings. All three Russian maturities show strong, statistically confident cointegration, with bounds F-statistics ranging from 6.33 to 8.78 and error-correction coefficients tightly clustered around −0.32 to −0.33 ($p < 0.001$ throughout). All three Kazakhstan maturities fail to reject the null of no cointegration under the joint bounds test.

A subtlety is worth flagging: Kazakhstan's individual error-correction coefficients approach or reach conventional significance as maturity increases (10yr: $p = 0.024$, technically significant on its own), yet the joint bounds F-test still indicates no cointegration even at 10yr. This is not a contradiction — the bounds test is a joint test across all lagged-level terms and is the authoritative verdict in the Pesaran-Shin-Smith framework; a single coefficient's own significance does not override it.

The pattern splits cleanly by *country*, not by maturity: Russia passes at all three maturities, Kazakhstan fails at all three. A genuine economic difference in underlying dynamics would more plausibly produce maturity-dependent variation within each country; a uniform pass/fail split that tracks sample size (109 vs. 27) almost exactly is the signature expected if the deciding factor is statistical power rather than a true structural difference between the two countries' pre-war credit markets.

4.4 Specification B: Minimal Kazakhstan, Full Russia, Pre-War

Table 4.3: Specification B results, pre-war sample

Country	Maturity	n	EC term (p)	Bounds F-stat	Verdict
KZ	1yr	27	−0.072 (0.568)	2.17	No cointegration
RU	1yr	74	−0.503 (<0.001)	5.31	Cointegration
KZ	5yr	27	−0.265 (0.068)	1.11	No cointegration
RU	5yr	74	−0.418 (<0.001)	2.83	Inconclusive
KZ	10yr	27	−0.302 (0.024)	1.99	No cointegration
RU	10yr	74	−0.306 (0.008)	1.51	No cointegration

Findings. Adding Russia's full regressor set (inflation, GDP, external debt/GDP, reserves/M2, volatility) reduces its usable pre-war sample from 109 to 74 observations, since several added variables — particularly the annualized fiscal ratios — have shorter coverage than RUONIA/oil/VIX alone.

Russia's cointegration result proves sensitive to this specification change: it holds cleanly only at 1yr; the 5yr result weakens to inconclusive, and the 10yr result weakens to no cointegration — the same verdict as Kazakhstan. This mirrors the same statistical-power mechanism documented for Kazakhstan throughout this chapter, now visible in Russia's results once its effective sample size and parameter count move closer to Kazakhstan's constrained regime. The strength of Russia's pre-war cointegration finding should therefore be reported as specification-dependent rather than uniformly robust; Specification A provides the strongest, most stable evidence across all three maturities.

Table 4.4: Specification C results, pre-war sample

Country	Maturity	n	EC term (p)	Bounds F-stat	Verdict
KZ	1yr	27	−0.669 (0.149)	1.18	No cointegration
RU	1yr	74	−0.503 (<0.001)	5.31	Cointegration
KZ	5yr	27	−0.091 (0.804)	1.50	No cointegration
RU	5yr	74	−0.418 (<0.001)	2.83	Inconclusive
KZ	10yr	27	−0.018 (0.941)	2.22	Inconclusive
RU	10yr	74	−0.306 (0.008)	1.51	No cointegration

4.5 Specification C: Full Model, Both Countries, Pre-War

Russia's results are identical to Specification B, since its full regressor set was already in use there.

Findings. Kazakhstan's full specification did not fail outright, but its estimates become progressively less meaningful as maturity increases: the error-correction coefficient's p-value rises from 0.149 (1yr) to 0.804 (5yr) to 0.941 (10yr) — by 10yr the coefficient carries essentially no statistical information. This is the empirical realization of the small-sample concern raised in Section 4.2: with 8 regressors and enforced lags estimated on only 27 observations, the model technically converges but produces estimates too imprecise to interpret, rather than genuine evidence of no relationship.

Synthesis across Specifications A–C. Kazakhstan fails to show cointegration under every specification tested, with results becoming less reliable — not more informative — as regressors are added, given its fixed 27-observation constraint. Russia shows robust cointegration only under the minimal specification across all maturities; its result weakens progressively at longer maturities once the full regressor set is introduced. Taken together, these results support reporting Specification A as the primary specification for both countries, with Specifications B and C retained as documented robustness checks rather than alternative preferred models.

4.6 Full-Sample Estimation with a War Dummy

To test whether the long-run relationship extends across the full sample once the war's potential effect on the equilibrium level is accounted for, Specification A was re-estimated on the full 1987–2026 sample (constrained by each variable's own coverage) with a binary war dummy added as a regressor.

Table 4.5: Specification A, full sample with war dummy

Country	Maturity	n	EC term (p)	war_dummy.L1 (p)	Verdict
KZ	1yr	82	-0.200 (0.001)	-0.033 (0.925)	Cointegration
KZ	5yr	82	-0.115 (0.039)	—	No cointegration
KZ	10yr	82	-0.149 (0.010)	—	No cointegration
RU	1yr	164	-0.334 (<0.001)	-0.074 (0.498)	Cointegration
RU	5yr	164	-0.201 (<0.001)	-0.238 (0.075)	Cointegration
RU	10yr	164	-0.164 (<0.001)	-0.256 (0.088)	Inconclusive

Key finding. Kazakhstan's 1yr spread shows confirmed cointegration here, in contrast to every pre-war-only specification (Sections 4.3–4.5), where it consistently failed. With the sample roughly tripling ($27 \rightarrow 82$ observations) via inclusion of the post-war period, this supports the interpretation that Kazakhstan's pre-war-only null results reflected insufficient statistical power rather than a genuine absence of long-run relationship. The maturity-dependent pattern persists, however: Kazakhstan's 5yr and 10yr spreads still fail to show cointegration even with the larger full sample, echoing the term-structure heterogeneity identified in the exploratory analysis (Chapter 3), where oil-decoupling and domestic-liquidity effects were found to strengthen specifically with maturity.

An important methodological correction. Across every model in Table 4.5, the war dummy itself is statistically insignificant (all $p \geq 0.075$). This means the cointegration detected in these full-sample models should *not* be interpreted as evidence that the war produced a detectable shift in the long-run equilibrium level — the dummy carries no independent explanatory power in this specification. Rather, both countries exhibit a genuine long-run relationship between spread and domestic-liquidity/oil/VIX across the *entire* sample period, largely independent of whether the war dummy is included.

This result is best read as evidence that **a simple constant-shift dummy is not an adequate tool for detecting the war's structural effect** within the ARDL bounds-testing framework — a finding that directly motivates the subsequent Markov-switching analysis (Chapter 5, which detects regime changes endogenously rather than assuming a known break date) and the VAR/IRF analysis (Chapter 6, which tests for changes in dynamic transmission rather than a simple level shift) as necessary complements, not redundant robustness checks.

Consequently, the pre-war-only results in Table 4.2 remain the most direct and interpretable test of H1's "pre-war similarity" claim: Russia shows robust pre-war cointegration at all three maturities; Kazakhstan's sample is too short to test this claim with confidence in isolation. The full-sample results in Table 4.5 are a separate finding — a long-run relationship exists across the full period for both countries — and should not be conflated with a war-driven shift specifically.

4.7 Robustness Check: Expanded Specification for Kazakhstan's 5yr and 10yr Spreads

Given Kazakhstan's persistent failure to show cointegration at longer maturities, four candidate variables — external debt/GDP, reserves/M2, inflation, and GDP — were each added individually (not jointly, to avoid repeating the overfitting problem documented in Section 4.5) to the full-sample, war-dummy specification.

Table 4.6: Expanded specification test, Kazakhstan 5yr/10yr

Maturity	Added variable	n	EC (p)	F-stat	Verdict
5yr	External debt/GDP	77	−0.108 (0.079)	1.22	No cointegration
5yr	Reserves/M2	80	−0.137 (0.020)	1.30	No cointegration
5yr	Inflation	81	−0.127 (0.073)	1.60	No cointegration
5yr	GDP	77	−0.088 (0.144)	1.37	No cointegration
10yr	External debt/GDP	77	−0.171 (0.006)	2.33	No cointegration
10yr	Reserves/M2	80	−0.165 (0.020)	2.05	No cointegration
10yr	Inflation	81	−0.132 (0.047)	1.64	No cointegration
10yr	GDP	77	−0.129 (0.042)	2.11	No cointegration

Findings. None of the four candidate variables recovers cointegration at either maturity, even though several individual error-correction coefficients reach conventional significance on their own (e.g., 10yr with external debt/GDP, $p = 0.006$). The joint bounds F-test remains decisively below the lower critical bound in every case, consistent with the coefficient-versus-joint-test distinction noted in Section 4.3.

This is treated as a genuine, informative null result rather than an unresolved gap. Kazakhstan's 5yr and 10yr spreads do not exhibit a detectable long-run relationship with domestic liquidity, global risk factors, or any of the fiscal/-macro variables tested here, even after accounting for the war via a level-shift dummy. Combined with the exploratory finding that oil-decoupling and liquidity-sensitivity effects were strongest specifically at the long end (Chapter 3), this suggests Kazakhstan's long-maturity spread may be governed by maturity-specific factors — market liquidity conditions, issuance-specific dynamics, or investor composition — that fall outside the macro-financial variable set examined in this paper. This is documented as a legitimate scope limitation rather than a modeling failure to be pursued further.

4.8 Interpreting the Long-Run Coefficients: Russia's Pre-War Baseline

Given Specification A's pre-war results (Table 4.2) are the most robust in this analysis, the underlying coefficients are interpreted in full as the baseline against which post-war transmission changes (Chapters 5–6) are compared.

Table 4.7: Russia, Specification A, pre-war: long-run coefficients

Variable	1yr coef. (p)	5yr coef. (p)	10yr coef. (p)
Error-correction (own lag)	−0.318 (<0.001)	−0.324 (<0.001)	−0.329 (<0.001)
RUONIA	0.027 (0.079)	0.061 (<0.001)	0.099 (<0.001)
Brent oil	0.0061 (0.003)	0.0024 (0.183)	0.0012 (0.478)
VIX	0.0204 (0.006)	0.0111 (0.120)	0.0078 (0.222)

The error-correction speed is remarkably stable across maturities (-0.318 to -0.329, all $p < 0.001$), implying that roughly 32% of any deviation from long-run equilibrium is corrected each month, for a half-life of approximately

1.8 months, regardless of maturity. This indicates one coherent equilibrium-reversion mechanism operating similarly across Russia's credit curve pre-war.

Domestic liquidity conditions (RUONIA) strengthen with maturity — the long-run coefficient roughly quadruples from 0.027 at 1yr (only marginally significant) to 0.099 at 10yr (highly significant), with precision increasing alongside magnitude. This is consistent with an economic interpretation in which short-term interbank rate movements are often transient (reflecting liquidity-management operations) and are therefore only partially priced into the shortest-maturity spread, while persistent liquidity conditions are more fully reflected in the compensation demanded for longer-duration credit risk.

Global risk factors (oil, VIX) show the opposite maturity pattern — both significant at 1yr (oil: $p = 0.003$; VIX: $p = 0.006$) but fading to insignificance at 5yr and 10yr. This suggests Russia's short-end spread was, pre-war, more exposed to immediate global risk-sentiment shocks, while the long end was more structurally anchored to domestic monetary conditions.

Oil's positive sign at 1yr (higher oil prices associated with a *higher* spread) is notable, since it runs counter to the "oil exporter, higher oil $\rightarrow$ lower credit risk" relationship documented for Kazakhstan. Given this effect fades to insignificance at longer maturities, it is best interpreted as a short-end-specific phenomenon — plausibly linked to capital-flow or ruble-volatility dynamics that coincided with oil-price movements within this particular pre-war sample — rather than a robust, curve-wide relationship between oil prices and Russian credit risk.

Overall interpretation. Russia's pre-war credit market exhibits a maturity-differentiated transmission mechanism: short-term spreads are more sentiment- and global-shock-driven, while long-term spreads are more structurally anchored to domestic monetary conditions. This coherent baseline is the reference point against which the post-war VAR/IRF analysis (Chapter 6) should assess whether — and how — this transmission mechanism changed.

4.9 Summary and Implications for Subsequent Analysis

1. **Method justification.** ARDL bounds testing was the appropriate choice given the confirmed mixed order of integration (Kazakhstan I(1), Russia I(0)) across the two countries' dependent variables, a setting in which Johansen or Engle-Granger cointegration testing cannot be validly applied.

2. **Specification choice.** The minimal specification (Section 4.3) is the most defensible primary result for both countries; fuller specifications (Sections 4.4–4.5) demonstrate the overfitting risk inherent in adding regressors against Kazakhstan's small pre-war sample, and reveal that Russia's own cointegration finding is not fully robust to specification choice at longer maturities.

3. **Pre-war cointegration (H1).** Russia shows robust, consistent cointegration across all three maturities pre-war. Kazakhstan's 27-observation pre-war sample does not permit an equally confident test — documented as a data limitation rather than evidence against a relationship, a conclusion strengthened by Kazakhstan's 1yr spread achieving cointegration once tested on its full sample (Section 4.6).

4. **The war dummy does not drive cointegration.** A corrected, fair comparison (Section 4.6) shows the war dummy is statistically insignificant across every model tested. A simple level-shift dummy is not an adequate tool for capturing the war's structural effect in this framework — motivating the Markov-switching and VAR/IRF analyses that follow.

5. **Kazakhstan's maturity-specific limitation.** The 5yr and 10yr spreads fail to show cointegration under every specification and every candidate additional variable tested (Section 4.7), suggesting their post-war dynamics are governed by factors outside this paper's macro-financial variable set — a legitimate scope limitation to state directly.

6. **Carrying forward.** Russia's stable pre-war error-correction speed and

maturity-differentiated sensitivity to domestic versus global factors (Section 4.8) form the baseline against which Chapter 6's post-war VAR/IRF results should be compared. Chapter 5's Markov-switching model should be evaluated for whether it detects a regime break near February 2022, given the war dummy's failure to capture a level shift here.

Chapter 5

Markov-Switching Analysis: Endogenous Regime Detection

5.1 Motivation and Methodology

This chapter uses Markov-switching models to test, endogenously, whether the Russia–Ukraine war produced a detectable regime shift in Kazakhstan's and Russia's credit spreads — without imposing February 2022 as the break date *a priori*. This directly addresses H2, and follows up on two open findings from the ARDL analysis (Chapter 4): (i) the war dummy included in the full-sample ARDL specifications was statistically insignificant in every model, indicating that a simple level-shift specification does not capture the war's structural effect; and (ii) the exploratory analysis (Chapter 3) identified a possible *second* shock episode in Russian spreads around late 2024–2025, which an assumed single break at February 2022 could not detect by construction.

Unlike a dummy variable, which assumes both the existence and the exact timing of a break, a Markov-switching model treats regime membership as a latent state $S_t \in \{0, 1, \ldots, k-1\}$, estimated directly from the data. In its simplest form, the model is

$$y_t = \mu_{S_t} + \varepsilon_t, \qquad \varepsilon_t \sim N\big(0,\, \sigma^2_{S_t}\big)\,, \tag{5.1}$$

where the regime S_t follows a first-order Markov chain with transition probabilities $p_{ij} = P(S_t = j \mid S_{t-1} = i)$. If the war produced a genuine structural break, this approach should recover a regime transition near February 2022 without being told to look for one — a stronger form of evidence than a dummy variable specified in advance to test exactly that hypothesis.

5.2 Data and Series Treatment

Consistent with the order-of-integration results established in Chapter 3, Kazakhstan's 1-year spread (I(1)) is modeled in first-differenced form, and Russia's 1-year spread (I(0)) is modeled in levels. This ensures each series has a well-defined, stable mean for the model to estimate within each regime, which a non-stationary level series would not provide.

5.3 Univariate Two-Regime Models

Two-regime models ($k = 2$) were estimated first as a baseline, with both the mean and variance permitted to switch by regime.

Table 5.1: Kazakhstan 1yr spread (differenced), two-regime model

Parameter	Regime 0 (Calm)	Regime 1 (Turbulent)
Mean	0.036 (0.385)	−0.017 (0.876)
Variance	0.040 (0.004)	0.471 (0.000)

p-values in parentheses.

$p[0 \to 0] = 0.820$, $p[1 \to 0] = 0.175$. $n = 81$, *Dec. 2019–Aug. 2026.*

Table 5.2: Russia 1yr spread (level), two-regime model

Parameter	Regime 0 (Low)	Regime 1 (High)
Mean	0.224 (0.052)	1.631 (0.000)
Variance	0.015 (0.450)	0.403 (0.000)

p-values in parentheses.

$p[0 \to 0] = 0.450$, $p[1 \to 0] = 0.004$. $n = 164$, *Jan. 2013–Aug. 2026.*

Neither of Kazakhstan's two regimes has a mean statistically distinguishable from zero; they are distinguished almost entirely by variance (Regime 1's variance is roughly ten times Regime 0's). This indicates the model detects *volatility clustering* in Kazakhstan's spread changes, not a persistent shift in the spread's level. Russia's two regimes, by contrast, differ in both mean and variance: Regime 1 represents a genuinely elevated credit-risk level (roughly seven times Regime 0's mean), not merely higher volatility. Regime 1 is additionally estimated as nearly absorbing ($p[1 \to 0] = 0.0044$, precisely estimated) — once Russia's spread enters this regime, the model estimates it essentially does not return to the low regime within the sample.

5.4 Locating the Regimes in Time

Smoothed regime probabilities were extracted and the exact calendar periods in which each regime's probability exceeds 0.5 were identified.

Kazakhstan's turbulent regime begins essentially at the war onset (February–May 2022) and recurs cyclically through 2026, consistent with the oscillating divergence pattern already identified in the exploratory rolling-correlation analysis (Chapter 3).

Russia's high-spread regime, however, does *not* align with February 2022 in the two-regime specification: it begins in January 2013 (near the start of the sample), briefly exits in early 2015, and then persists continuously from February 2015 through the end of the sample (August 2026) — more than eleven years without exit. Because Regime 1 is estimated as nearly absorbing, the war onset is absorbed into an already-ongoing elevated regime (plausibly dating to the 2014 oil-price collapse and the sanctions following Crimea's annexation) rather than triggering a separately identifiable transition. This is a limitation of the two-regime specification rather than evidence the war had no effect, and motivates the three-regime extension in Section 5.6.

5.5 A Formal Test of the Kazakhstan–Russia Relationship

To test directly whether the statistical relationship between the two countries' spreads changed after the invasion, a regime-switching regression was estimated with Russia's spread as the dependent variable and Kazakhstan's spread as a regressor, allowing the intercept, slope, and variance to switch by regime:

$$\mathrm{ru}_t = \alpha_{S_t} + \beta_{S_t}\,\mathrm{kz}_t + \varepsilon_t, \qquad \varepsilon_t \sim N\big(0,\, \sigma^2_{S_t}\big)\,. \tag{5.2}$$

Regime 0 shows a statistically significant relationship between the two spreads (slope -0.337, $p < 0.001$, tight residual variance). Regime 1 shows this relationship vanish entirely (slope statistically indistinguishable from zero, $p = 0.701$)

Table 5.3: KZ→RU relationship, two-regime model

Parameter	Regime 0	Regime 1
Intercept (α)	0.531 (0.000)	1.798 (0.000)
Slope on KZ (β)	−0.337 (0.000)	−0.056 (0.701)
Variance	0.0009 (0.270)	0.436 (0.000)

p-values in parentheses.

$p[1 \rightarrow 0] = 0.022$. $n = 81$, Dec. 2019–Aug. 2026.

alongside an approximately 480-fold increase in residual variance. This is direct, formally tested evidence that a genuine statistical relationship between the two countries' spreads exists in one regime and is lost in another — a stronger form of evidence for divergence than the descriptive rolling-correlation analysis alone provides. One specification caveat applies: Kazakhstan enters as a differenced series and Russia as a level (matching each country's confirmed order of integration), so the coefficient reflects a mixed-transform relationship; the qualitative finding is expected to be robust to this choice, though the exact magnitude should be read with this caveat in mind.

5.6 Three-Regime Extension: Russia

Given the two-regime model's inability to separate the 2015 shift from the 2022 war, a three-regime specification was tested for Russia's univariate series.

Table 5.4: Russia 1yr spread (level), three-regime model

Parameter	Regime 0 (Low)	Regime 1 (Moderate)	Regime 2 (Crisis)
Mean	0.219 (0.010)	1.143 (0.000)	2.103 (0.000)
Variance	0.014 (0.368)	0.096 (0.000)	0.233 (0.000)

p-values in parentheses. AIC = 227.2, BIC = 264.4 (two-regime: AIC = 332.7, BIC = 351.3).

Both information criteria improve substantially with three regimes, well beyond what added parameters alone would produce, supporting genuine three-state structure. Table 5.5 reports the resulting regime timeline.

The three-regime model resolves the central limitation of Section 5.4: the war (March 2022–August 2023, approximately 17 months) is now identified as a distinct crisis episode, separate from the 2015 shock, the COVID-19 shock, and a further crisis regime beginning October 2024. This provides methodologically

Table 5.5: Russia three-regime timeline

Period	Regime	Interpretation
2013-01 – 2014-12	Crisis	Pre-sample baseline
2014-12 – 2015-02	Low	Brief transition
2015-02 – 2015-12	Crisis	Post-Crimea / oil-price collapse
2015-12 – 2020-03	Moderate	Post-2014 “new normal”
2020-03 – 2021-02	Crisis	COVID-19 shock
2021-02 – 2022-03	Moderate	Brief de-escalation, pre-war
2022-03 – 2023-08	**Crisis**	**War (separately identified)**
2023-08 – 2024-10	Moderate	Partial de-escalation
2024-10 – 2026-08	**Crisis**	**Second shock episode**

independent confirmation that (i) the war constitutes a genuine, endogenously detected structural break, supporting H2, and (ii) a second, statistically distinct crisis regime emerges in late 2024 — corroborating the exploratory finding (Chapter 3) linked to Russia's documented fiscal and monetary deterioration, with short-term government bond yields reaching an all-time high of 22.78% in December 2024.

5.7 Three-Regime Extension: Kazakhstan (Rejected)

A matching three-regime model was tested for Kazakhstan and rejected: both AIC (120.1 vs. 114.4) and BIC (148.8 vs. 128.7) are worse than the two-regime specification. Two of the three estimated regimes have means statistically indistinguishable from zero and from each other, differing mainly in variance — the same distinction the two-regime model already captures — and several transition-probability standard errors could not be estimated, consistent with the model straining against Kazakhstan's smaller sample ($n = 81$). Unlike Russia, Kazakhstan's data does not support a three-regime characterization; the two-regime volatility-clustering model (Section 5.3) remains the appropriate specification. This asymmetry is attributed to Kazakhstan's shorter data history rather than a substantive economic difference in regime complexity between the two countries.

5.8 Refining the Relationship Model: Three Regimes

Given Russia clearly supports three regimes while Kazakhstan does not, a three-regime version of the relationship model (Equation 5.2) was tested directly, rather than assumed from either univariate result.

Table 5.6: KZ→RU relationship, three-regime model

Parameter	Regime 0	Regime 1	Regime 2
Intercept (α)	0.531 (0.000)	1.208 (0.000)	2.176 (0.000)
Slope on KZ (β)	−0.332 (0.000)	−0.188 (0.264)	−0.007 (0.955)
Variance	0.0009 (0.217)	0.122 (0.000)	0.268 (0.000)

p-values in parentheses. AIC $= 151.6$*, BIC* $= 187.5$ *(two-regime: AIC* $= 180.1$*, BIC* $= 199.3$*).*

Table 5.7: KZ→RU relationship, three-regime timeline

Period	Regime
2022-01 – 2022-03	**Relationship present**
2022-03 – 2023-07	No relationship, high RU spread
2023-09 – 2023-10	**Relationship present**
2023-10 – 2024-10	No relationship, moderate RU spread
2024-10 – 2026-08	No relationship, high RU spread

Only Regime 0 shows a statistically significant relationship between the two countries' spreads; Regimes 1 and 2 do not, differing from each other primarily in Russia's own spread level (moderate versus crisis, matching its univariate regimes). The significant-relationship regime occurs only in two narrow windows — January–March 2022 and September–October 2023 — while the no-relationship, high-spread regime dominates March 2022–July 2023 and again from October 2024 onward, closely tracking Russia's own crisis-regime dating (Table 5.5).

This refines the divergence narrative beyond a simple, sustained breakdown in comovement: the evidence instead supports *transient shock transmission at the moment of onset*, in which both countries' spreads move together briefly as a shock begins, followed by decoupling as each country's subsequent crisis dynamics evolve independently, driven by country-specific rather than shared factors.

5.9 Summary and Implications for the VAR/IRF Analysis

1. **Method justification.** Markov-switching models were used to test for a regime shift endogenously, directly motivated by the ARDL war dummy's failure to register statistical significance (Chapter 4).

2. **Kazakhstan.** A two-regime, volatility-driven structure is the best-supported specification; the turbulent regime aligns with the war onset and recurs cyclically thereafter, consistent with Chapter 3's rolling-correlation findings.

3. **Russia.** A three-regime, level-and-volatility structure is strongly supported and separately identifies the war (March 2022–August 2023) from the 2015 shock, the COVID-19 shock, and a second crisis episode beginning October 2024 — independently corroborating the second-shock finding from Chapter 3.

4. **The KZ–RU relationship.** A formally tested regime-switching regression shows the relationship between the two countries' spreads is significant only in brief onset windows (January–March 2022; September–October 2023), and absent for the remainder of each crisis period — refining the paper's divergence claim into a more specific transient-transmission-then-decoupling pattern.

5. **Implications for Chapter 6.** Russia's precisely dated crisis regimes provide sharper windows for structural-break analysis in the VAR specification than the assumed February 2022 date alone. The onset-versus-duration distinction found here suggests the VAR/IRF analysis should examine impulse responses in short windows around each shock's onset rather than assuming a uniform post-war transmission structure across the full 2022–2026 period.

Chapter 6

VAR/IRF Analysis: Oil Shock Transmission Before and After the War

6.1 Motivation and Specification

This chapter estimates a minimal vector autoregression (VAR) for each country's 10-year credit spread, domestic interbank liquidity rate, and Brent crude oil, to test whether the dynamic transmission of an oil shock to the spread changed after the war — directly following up on the oil-decoupling finding from Chapter 3 and the ARDL results of Chapter 4.

Unlike ARDL, which designates one variable as dependent, a VAR treats every variable in the system as jointly determined by the recent past of all variables in the system. With three variables $y_t = (\text{spread}_t, \text{liquidity}_t, \text{oil}_t)'$, a VAR(1) — one lag of each variable — is

$$y_t = c + Ay_{t-1} + \varepsilon_t, \tag{6.1}$$

where A is a 3×3 coefficient matrix estimated jointly across all three equations. The impulse response function (IRF) traces the effect of a one-time, unanticipated shock to one variable (here, oil) through the estimated system over subsequent periods, holding all else as the model implies.

The system was deliberately kept minimal — three variables, one lag, maturity restricted to 10 years — given Kazakhstan's constrained pre-war sample (26 observations after accounting for the lag). A richer specification would repeat the overfitting problem already documented extensively in Chapter 4.

6.2 Pre-War vs. Post-War: Full-Period Comparison

VAR(1) was estimated separately for the pre-war and post-war samples, for Kazakhstan (spread, TONIA, oil; pre-war $n = 26$, post-war $n = 55$) and Russia (spread, RUONIA, oil; pre-war $n = 109$, post-war $n = 55$).

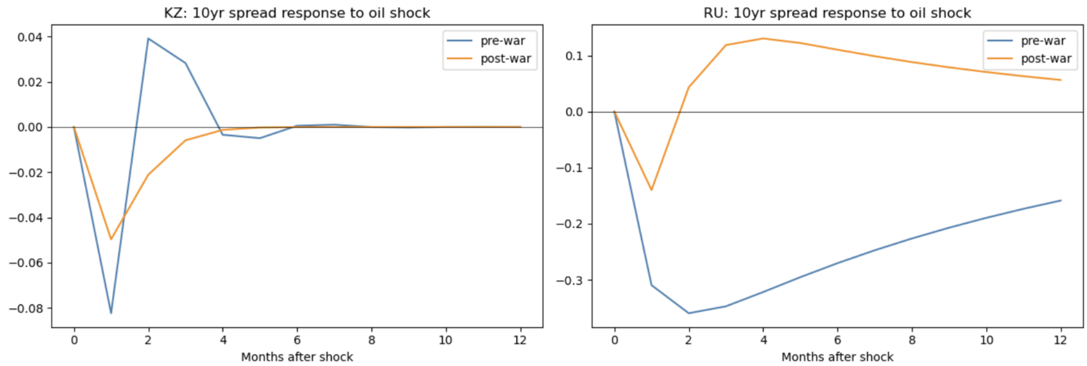

Figure 6.1: Response of the 10-year credit spread to a one-standard-deviation oil shock, pre-war vs. post-war, Kazakhstan and Russia.

Kazakhstan. The pre-war response is sharp and short-lived: the spread falls to approximately -0.08 by month 1, rebounds to roughly $+0.04$ by month 2 (an overshoot), then decays to zero by month 6. The post-war response is smaller in magnitude (bottoming near -0.05) and monotonic, with no overshoot, fading to zero over a similar horizon. The oil-spread relationship is dampened post-war but not eliminated — consistent with, though more moderate than, the near-total correlation breakdown found in the static analysis of Chapter 3.

Russia. Pre-war, an oil shock produces a deep and slowly-decaying negative response, reaching approximately -0.36 by month 2 and still around -0.16 at month 12 — not fully reverted within the horizon shown. The single post-war VAR, spanning the entire February 2022 to August 2026 window, shows a much smaller initial response that reverses sign by month 3 and settles into a modest positive plateau ($+0.06$ to $+0.13$) for the remainder of the horizon.

6.3 A Necessary Correction: Sub-Regime Decomposition

The single post-war VAR result for Russia (Section 6.2) should not be read as a stable new transmission regime. Chapter 5 identified, via an endogenously-estimated three-regime Markov-switching model, two statistically distinct crisis episodes within the post-war window for Russia: the war regime itself (March 2022 to August 2023) and a separate, later shock (October 2024 onward). Chapter 5's own conclusion cautioned against treating the full post-war period as one uniform transmission structure; Russia's post-war VAR was therefore re-estimated separately within each of these two Markov-dated windows ($n = 18$ and $n = 23$ respectively).

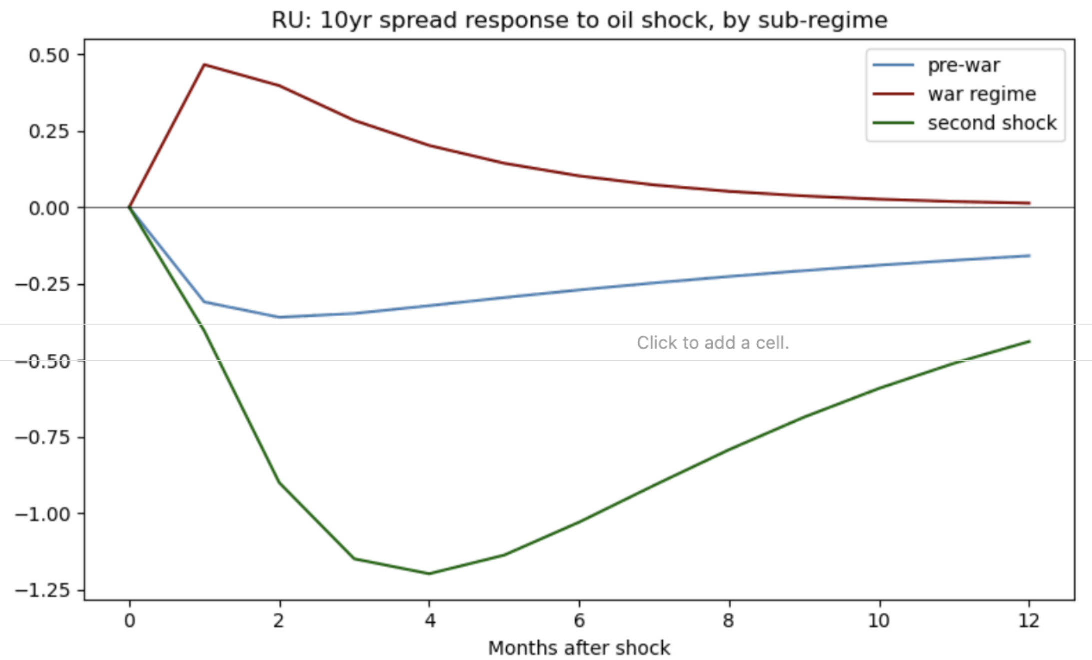

Figure 6.2: Response of Russia's 10-year credit spread to an oil shock: pre-war, the war regime (March 2022–August 2023), and the second-shock regime (October 2024–2026).

The sub-regime split reveals that the single-block post-war result (Section 6.2) was an averaging artifact masking two genuinely different, opposite-signed dynamics. **During the war regime**, an oil shock *raises* the spread — a sign reversal from the pre-war relationship — peaking near +0.47 at month 1 before decaying toward zero by month 12. **During the second-shock regime**, the response is same-signed as pre-war but far more extreme: the spread falls to approximately -1.2 by month 4, more than three times the pre-war trough, and remains substantially depressed (-0.45) even after 12 months.

Revised conclusion. The war did not produce a single new, stable oil-transmission mechanism for Russia's credit spread. It produced at least two distinct and opposite-signed dynamics, corresponding to the two crisis regimes independently identified in Chapter 5. The single-block post-war estimate in Section 6.2 is retained here only to illustrate why this decomposition was necessary, not as a standalone finding.

6.4 Scope Limitation

Chapter 5 additionally suggested that Kazakhstan's volatility-driven (rather than level-driven) regime structure might warrant a VAR specification that explicitly models conditional volatility alongside the mean equation — for example, a VAR-GARCH extension. This was considered and deliberately not pursued here: it represents a substantially larger modeling undertaking than the constant-variance VAR used in this chapter, and the marginal benefit for Kazakhstan's already-thin sample does not justify the added complexity within the scope of this thesis. This is noted explicitly as a natural extension for future work, rather than omitted silently.

6.5 Summary

A minimal three-variable VAR, estimated separately by period and (for Russia) by Markov-dated sub-regime, shows that the oil-to-spread transmission mechanism changed after the invasion in both countries, but in different ways. Kazakhstan's response weakened and lost its short-run overshoot while remaining directionally consistent with the pre-war relationship. Russia's response did not settle into one new pattern: it reversed sign during the war itself, then reverted to the original sign at far greater magnitude during the second shock identified in Chapter 5.

Both results reinforce the conclusion, established across Chapters 4 and 5, that a single before/after framing is insufficient to characterize how the war altered credit-risk transmission in either market.

Chapter 7

Concluding Remarks

7.1 Summary

This thesis asked whether the Russia-Ukraine war changed how corporate credit-risk spreads respond to macro-financial conditions in Russia and Kazakhstan, and whether that change caused the two countries' credit markets to diverge. Three hypotheses structured the investigation: that the two countries' spreads responded similarly to common determinants before the war (H1); that the war produced a structural break in this relationship, detectable independently of any assumed break date (H2); and that the two countries diverged in how they absorbed the shock (H3). Four empirical chapters, each building directly on a limitation exposed by the one before it, addressed these in turn.

H1 (pre-war similarity). Chapter 4's ARDL bounds tests found Russia's pre-war credit spread cointegrated robustly with domestic liquidity, oil, and VIX at all three maturities, with a stable error-correction speed ($\sim$32% monthly) and a maturity-differentiated sensitivity to global versus domestic drivers. Kazakhstan's markedly shorter pre-war sample (27 observations against Russia's 109) did not permit an equally confident test, and this was documented as a data limitation throughout rather than treated as evidence against the hypothesis — a conclusion strengthened by Kazakhstan's 1-year spread achieving cointegration once tested on its full sample. H1 is therefore best read as supported for Russia directly, and plausible but unconfirmed for Kazakhstan given the data available.

H2 (an endogenously detectable break). A simple war dummy, tested directly within the ARDL framework, was statistically insignificant in every specification for both countries — a level-shift dummy could not capture the war's effect. Chapter 5's Markov-switching models succeeded where the dummy did not: Kazakhstan's spread showed a volatility-driven regime shift aligning closely with the war onset, and Russia's three-regime model separately and precisely dated a war regime (March 2022 to August 2023). H2 is supported, but with an important qualification the dummy-based approach could never have revealed: the break Markov-switching detects is not a simple level shift, and for Russia it is not the only break in the post-war sample — a second, independent crisis regime emerges in October 2024, corroborated across Chapters 3, 5, and 6.

H3 (post-war divergence). The evidence for divergence is real but more intricate than a single before/after contrast. Kazakhstan's oil-decoupling and liquidity-strengthening effects (Chapter 3) strengthen with maturity rather than applying uniformly, and its 5- and 10-year spreads never achieve ARDL cointegration under any specification tested, including an expanded fiscal variable set — a genuine scope limitation rather than an unresolved gap. The rolling correlation between the two countries' spreads (Chapter 3) drops sharply negative at the war onset but oscillates rather than settling into a new stable state, and a formally estimated regime-switching relationship (Chapter 5) shows the two spreads are significantly related only in brief windows coinciding with shock onsets, not on a sustained basis. Chapter 6's impulse responses sharpen this further: Russia's oil-to-spread transmission reverses sign during the war itself and reverses again, at far greater magnitude, during the second shock — evidence against a single new post-war regime and for a shock-specific, non-stationary transmission mechanism.

Taken together, the war changed the relationship between credit spreads and macro-financial conditions in both countries, and this change differed between them — but the divergence is not a one-time before/after shift. It is characterized by different transmission channels in each country, an independently-discovered second shock tied to Russia's sustained fiscal deterioration rather than the invasion alone, and cross-country comovement that reconnects briefly at moments of acute shock before decoupling again.

Methodologically, each chapter's limitation motivated the next chapter's approach rather than being worked around within the same framework: ARDL established the mixed order of integration and exposed the war dummy's failure; Markov-switching supplied the endogenous break detection ARDL could not, and surfaced a second regime the original hypotheses did not anticipate; VAR/IRF, informed by the Markov-dated regimes rather than an assumed uniform post-war window, revealed dynamics a single-block estimate would have averaged away. Kazakhstan's constrained data history is the central limitation running through every chapter, and is stated plainly rather than overstated or hidden at each point it binds.

7.2 Future Directions

Several extensions follow directly from limitations acknowledged in the preceding chapters, rather than being introduced here for the first time.

A longer Kazakhstan sample. Kazakhstan's government bond yield curve data begins only in November 2019; as this series lengthens with time, the ARDL and Markov-switching results for Kazakhstan — particularly the untested 5- and 10-year cointegration relationships (Chapter 4) — could be re-examined with substantially more statistical power.

Conditional-volatility modeling for Kazakhstan. Chapter 5 found Kazakhstan's regime structure to be volatility-driven rather than level-driven, and Chapter 6 noted, without pursuing, that a VAR-GARCH specification might capture this more directly than the constant-variance VAR used here. This remains a natural and well-motivated extension.

Higher-frequency volatility data. The volatility measure used throughout (Chapter 2) is constructed from monthly stock index returns; realized volatility computed from daily or intraday data, if obtained, would be a materially sharper proxy.

Formal structural-break testing within the VAR itself. The sub-regime decomposition in Chapter 6 relied on break dates estimated separately by the univariate Markov-switching models of Chapter 5. A multivariate structural-break test applied directly to the VAR system (e.g., a Bai-Perron-type procedure) would allow the break dates and the transmission-mechanism estimates to be determined jointly, rather than in two separate steps.

Extension beyond the Kazakhstan-Russia pair. Whether the patterns found here — particularly Kazakhstan's transient, onset-concentrated comovement with Russia (Chapter 5) — are specific to this pair or generalize to other economies with close financial or trade ties to Russia (other Central Asian or CIS economies) is an open question this thesis's two-country design cannot address on its own.

Additional fiscal variables. A current account-to-GDP ratio was considered during the data construction stage (Chapter 1) but not included, as a suitably clean monthly or reliably disaggregated series could not be sourced for both countries within the scope of this thesis. Incorporating it would extend the fiscal ratio analysis of Chapter 2 beyond external debt and reserves.

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APPENDICES

Appendix A

Full Data Dictionary

Table A.1 reproduces the data dictionary of Chapter 1 (Table 1.1) with full source URLs included, for auditability.

Table A.1: Full data dictionary with source URLs

ID	Variable	Source URL
1	kz_corp_yield_m	https://kase.kz/ru/indexes-and-indicators/bonds/
2	ru_corp_yield_m	https://moex.com/en/index/RUCBITR/archive
3–5	kz_govt_yield_{1,5,10}y_m	https://kase.kz/en/markets/markets-valuation/yield-curve-parameters
6–8	ru_govt_yield_{1,5,10}y_m	https://www.moex.com/download/zcyc/zcyc_parameters_archive.csv.zip
9	kz_tonia_m	https://kase.kz/en/indexes-and-indicators/repo/tonia
10	kz_usdfx_m	https://nationalbank.kz/en/exchangerates/
11	kz_inflation_yoy_m	https://taldau.stat.gov.kz/en/PivotGrid/PivotTable
12	kz_gdp_m	https://taldau.stat.gov.kz/en/
13	kz_extdebt_m	https://data.nationalbank.kz/statistics
14	kz_reserves_m	https://nationalbank.kz/en/international-reserve-and-asset/
15	kz_m2_m	https://nationalbank.kz/en/
16	kz_indprod_m	https://taldau.stat.gov.kz/en/PivotGrid/PivotTable
17	kz_unemp_m	https://taldau.stat.gov.kz/en/PivotGrid/PivotTable
18	kz_stockidx_m	https://kase.kz/en/indexes-and-indicators/shares/kase-index
19	ru_ruonia_m	https://www.cbr.ru/eng/hd_base/ruonia/dynamics/
20	ru_usdfx_m	https://www.cbr.ru/eng/currency_base/dynamics/
21	ru_inflation_yoy_m	https://www.rateinflation.com/inflation-rate/russia-historical-inflation-rate/
22	ru_gdp_m	https://fred.stlouisfed.org/series/NGDPRNSAXDCRUQ
23	ru_extdebt_m	https://www.cbr.ru/eng/statistics/macro_itm/external_sector/pb/p_balance/
24	ru_reserves_m	https://www.cbr.ru/eng/hd_base/mrrf/mrrf_7d/
25	ru_m2_m	https://www.cbr.ru/eng/statistics/macro_itm/dkfs/monetary_agg/
26	ru_indprod_m	https://eng.rosstat.gov.ru/folder/160366
27	ru_unemp_m	https://eng.rosstat.gov.ru/folder/160366
28	ru_stockidx_m	https://www.moex.com/en
29	gl_oil_brent_m	https://fred.stlouisfed.org/series/DCOILBRENTEU
30	gl_gold_m	https://www.investing.com/commodities/gold-historical-data
31	us_yield_1y_m	https://fred.stlouisfed.org/series/DGS1
32	us_yield_5y_m	https://fred.stlouisfed.org/series/DGS5
33	us_yield_10y_m	https://fred.stlouisfed.org/series/DGS10
34	us_fedfunds_m	https://fred.stlouisfed.org/series/FEDFUNDS
35	gl_vix_m	CBOE (no persistent series URL)

Appendix B

Full Stationarity Diagnostics

Chapter 2 (Section 2.3) reports the applied transform and final ADF/KPSS statistics for all 35 base variables in summary table form. Figures B.1–B.7 below show the corresponding raw-level and transformed series for every variable, for readers who wish to inspect the underlying diagnostic plots directly rather than rely on the summary statistics alone.

Figure B.1: Raw level and transformed series: `kz_corp_yield_m`, `ru_corp_yield_m`, `kz_govt_yield_1y_m`.

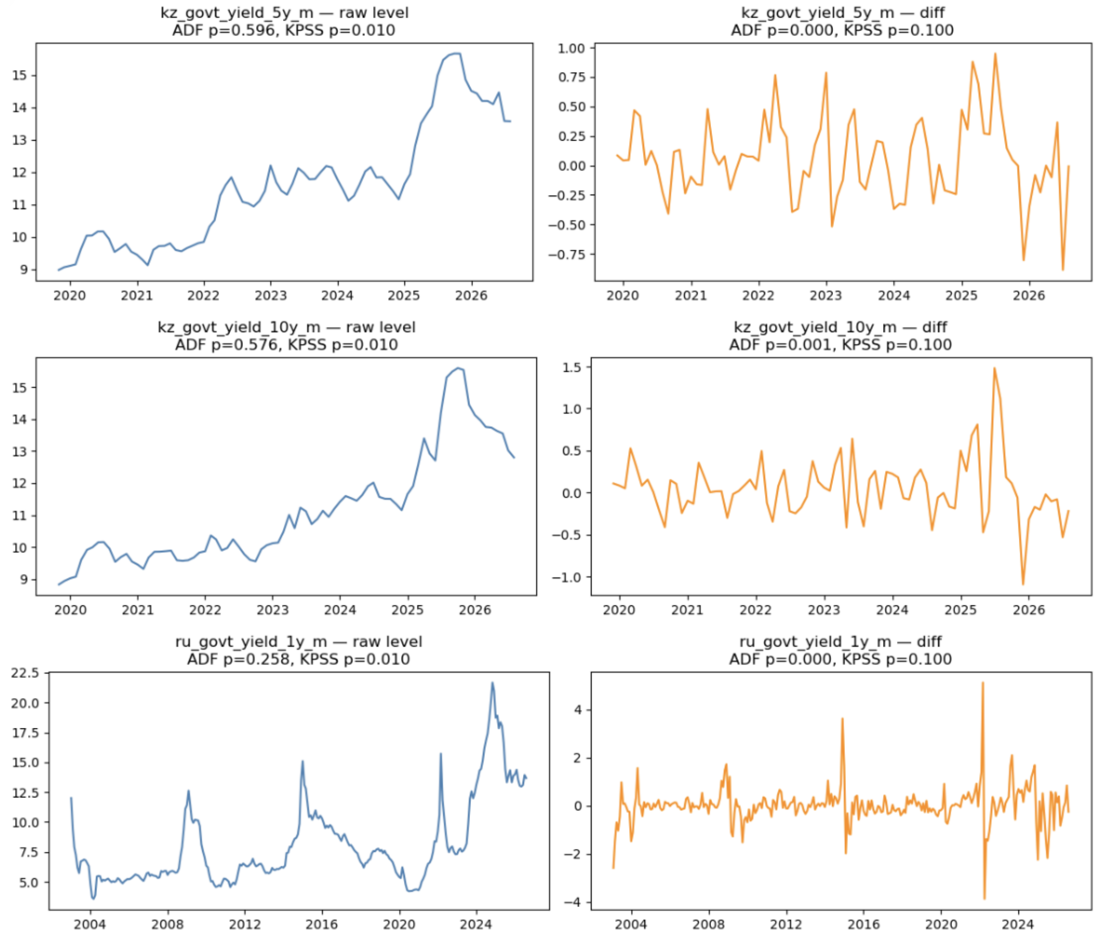

Figure B.2: Raw level and transformed series: `kz_govt_yield_5y_m`, `kz_govt_yield_10y_m`, `ru_govt_yield_1y_m`.

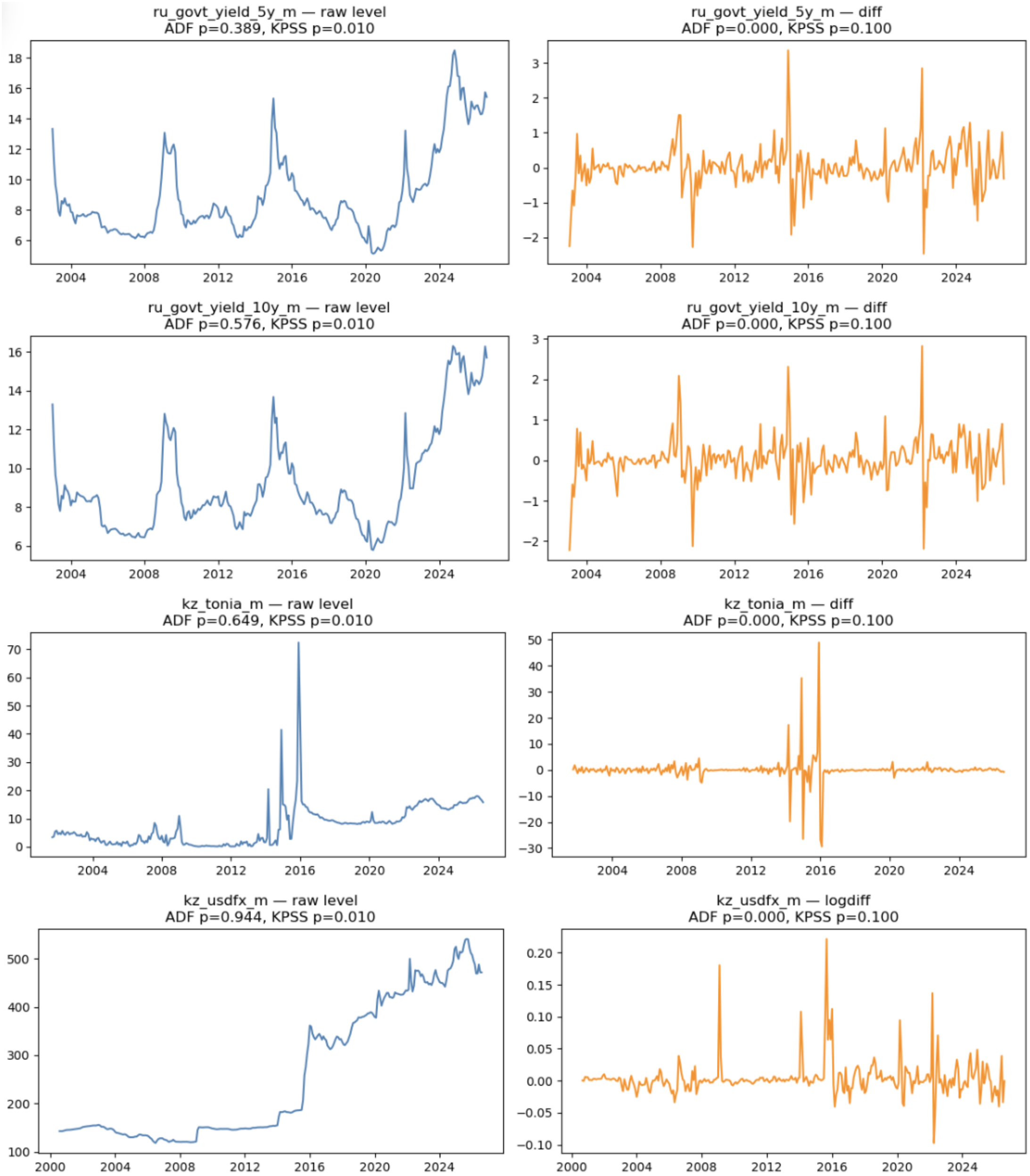

Figure B.3: Raw level and transformed series: `ru_govt_yield_5y_m`, `ru_govt_yield_10y_m`, `kz_tonia_m`, `kz_usdfx_m`.

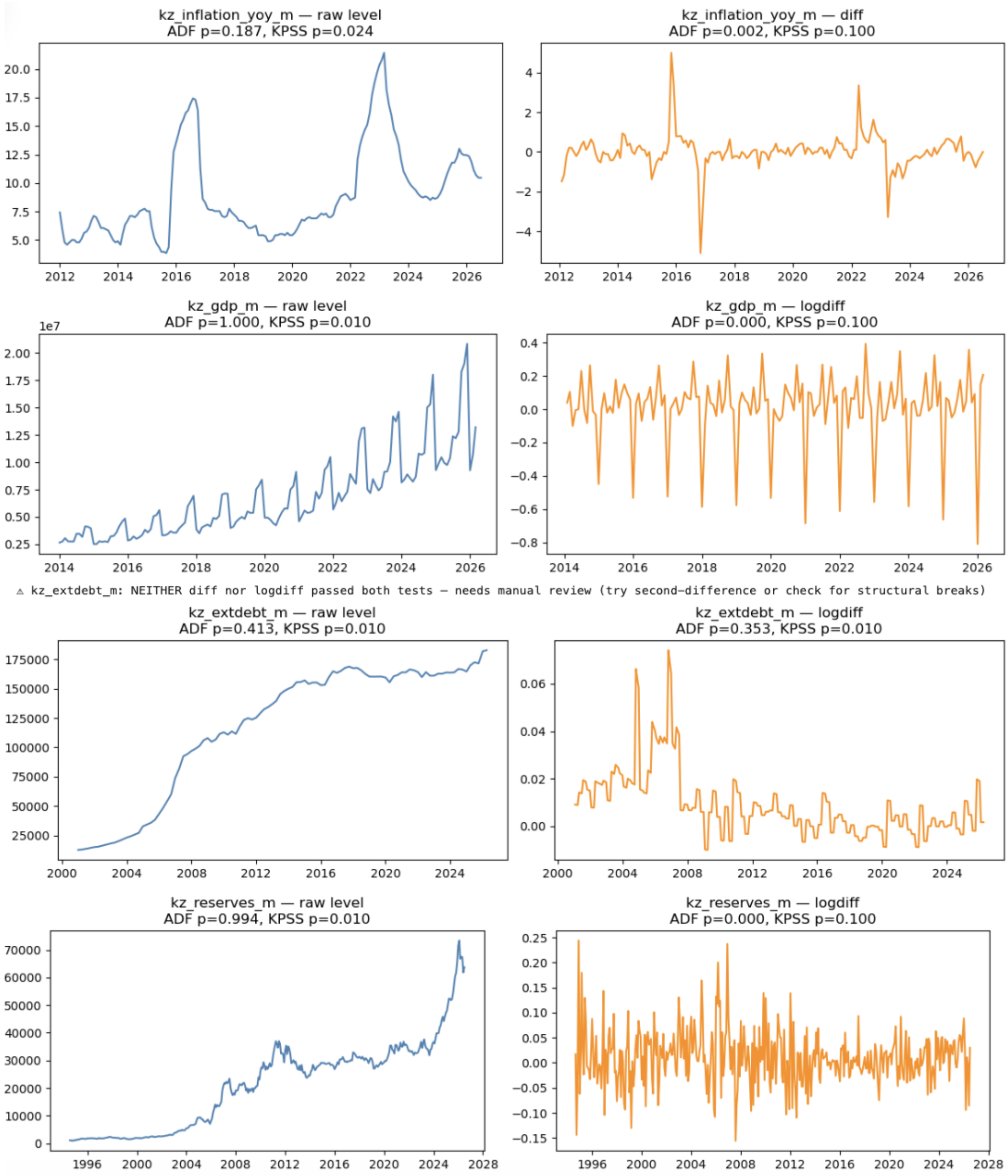

Figure B.4: Raw level and transformed series: kz_inflation_yoy_m, kz_gdp_m, kz_extdebt_m (first-order transform insufficient; see Chapter 2, Section 2.4), kz_reserves_m.

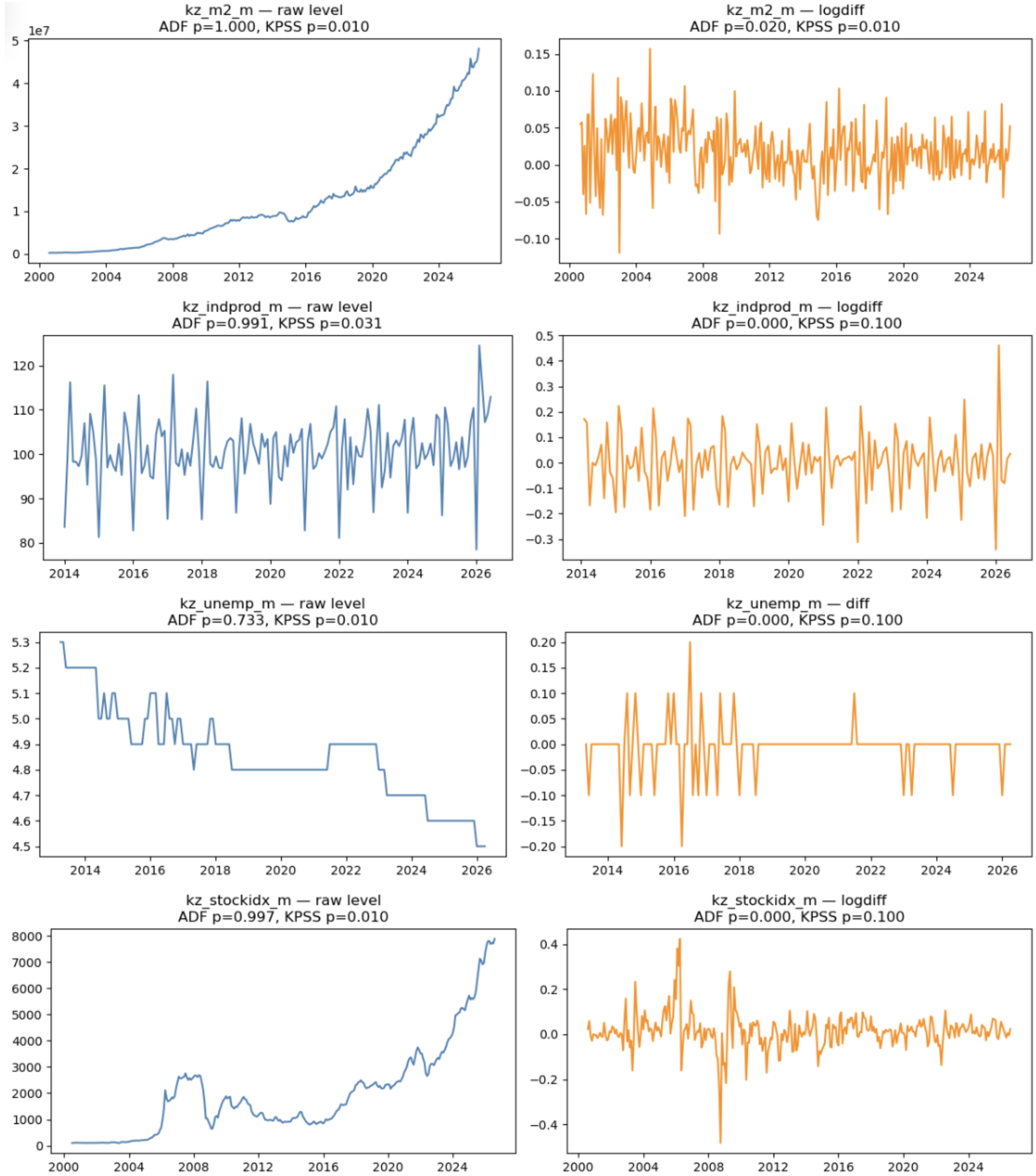

Figure B.5: Raw level and transformed series: `kz_m2_m` (first-order transform insufficient; see Chapter 2, Section 2.4), `kz_indprod_m`, `kz_unemp_m`, `kz_stockidx_m`.

Figure B.6: Raw level and transformed series: `ru_ruonia_m`, `ru_usdfx_m`, `ru_inflation_yoy_m`, `ru_gdp_m`.

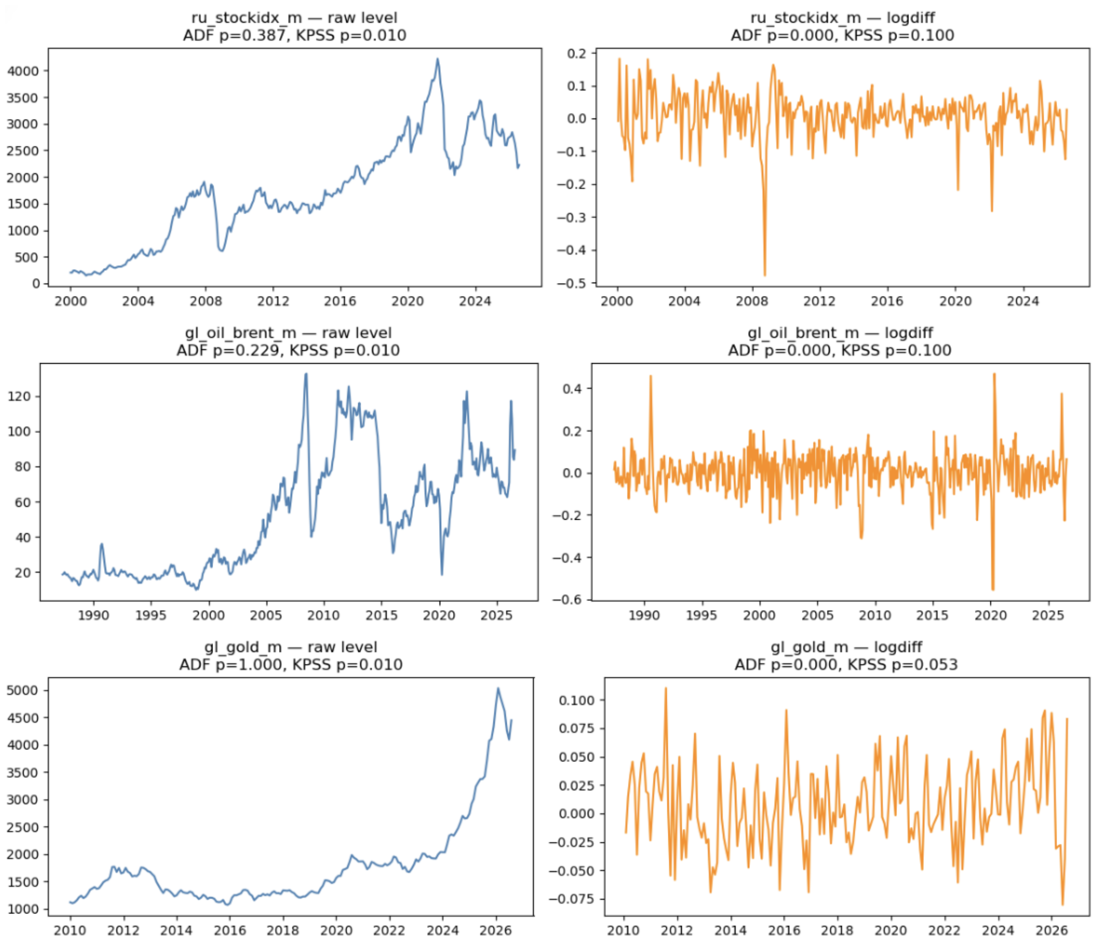

Figure B.7: Raw level and transformed series: `ru_stockidx_m`, `gl_oil_brent_m`, `gl_gold_m`.

Appendix C

Code and Data Availability

The complete analysis pipeline — data collection and cleaning, the seven Jupyter notebooks corresponding to Chapters 1 through 6, and the final merged dataset — is available at:

`https://github.com/darkhanmashirapov/Writing_Sample_Fall2027.git`

The repository is organized to mirror the chapter structure of this thesis:

- `01_data_collection.ipynb` — source-file loading and the four intermediate merged files (Chapter 1)
- `02_data_cleaning.ipynb` — master dataset construction, stationarity testing, and variable construction (Chapter 2)
- `03_eda.ipynb` — exploratory analysis (Chapter 3)
- `04_ardl.ipynb` — ARDL bounds testing (Chapter 4)
- `05_markov_switching.ipynb` — regime-switching analysis (Chapter 5)
- `06_var_irf.ipynb` — VAR/IRF analysis (Chapter 6)
- `master_dataset_final.xlsx` — the final dataset (Levels, Transformed, and Codebook sheets), as constructed in Chapter 2

Notebooks should be run in corresponding order to reproduce all results reported in this thesis.